

Market Concentration and Aggregate Productivity: The Role of Demand

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Abstract

In the typical market, a few large firms hold their positions through advantages in productivity and in demand. Grounded in new facts on firms' marketing expenditures, we build a model with oligopolistic firms that invest in demand to complement their productivity and entrench their market power. Because markups rise with market share, the return to marketing peaks for firms with significant pre-existing customer capital. This can create a *legacy effect* where unproductive firms retain large market share through investments in demand. "Good" concentration is productivity-driven; "bad" demand-driven concentration can persist due to the legacy effect. Aggregate productivity is about 5% below the level a planner reaches by correcting the strategic incentive. Standard size-based policies can backfire by entrenching incumbents.

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1 Introduction

Competition in the typical market is competition among a few large firms, each large enough to move market aggregates. These firms hold their positions on two margins, improving productivity and investing in demand, and they spend more on the second: sales and marketing expenditures average 4.2% of revenue against 2.4% for R&D among publicly-traded firms (He et al., 2024). Productivity sits at the center of the modern framework of strategic firm behavior (Atkeson and Burstein, 2008; Edmond et al., 2023); demand investment does not, even with high marketing spending and as concentration and market power draw growing attention (De Loecker et al., 2020; Olmstead-Rumsey, 2022; Akcigit and Ates, 2023; De Ridder, 2024).

This paper studies strategic demand investment under oligopoly and traces its consequences for market concentration and aggregate productivity. We capitalize firm-level marketing expenditure into a stock of customer capital and measure it alongside productivity for publicly traded U.S. firms. We ask the following questions: Is demand investment efficient or distortionary? How does demand investment affect concentration and aggregate productivity? How do size-based policies interact with efficiency when this channel is taken into account? We start with three facts to motivate the theoretical and quantitative framework.

Fact 1 (Customer capital tracks firm size more closely than productivity does).

Within industries, customer capital correlates with revenue around 0.75, while labor productivity and revenue are correlated around 0.50.

Fact 2 (Customer capital is more persistent than productivity, and its investment is path dependent). A simple AR(1) is 0.98 against 0.86 for labor productivity, and marketing investment growth is positively autocorrelated.¹

Fact 3 (Markets are granular). A typical market has 4–7 effective competitors, across both publicly traded and consumer-packaged goods firms.²

In a standard firm dynamics model, productivity and size should have a one-to-one correspondence. A long literature interprets the gap between size and productivity as

¹David and Venkateswaran (2019) report negative growth autocorrelation for physical capital. The positive sign for customer-capital investment motivates the path dependency discussed in Section 3.

²We use the sales-weighted inverse Herfindahl index across various market definitions.

allocative inefficiency, the output lost when inputs do not flow to their most productive use (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). Customer capital also may speak to this reading: on its own it explains a significant portion of firm size (Fact 1), and its path-dependent accumulation (Fact 2) suggests dynamics are essential for misallocation. Whether the overall investment in marketing is productive turns on the incentives of the handful of strategic firms who direct the flow of demand (Fact 3).

In our model, granular firms compete in a Cournot environment under CES demand, differ in productivity and customer capital, and invest in marketing for two reasons: to complement their productivity (the *productivity incentive*) and to capture the market power that comes from markups that increase in size (the *size incentive*). We microfound demand with rational inattention. Because the granular dynamic game is otherwise intractable, we solve it globally with a new computational method that scales to empirically relevant firm counts. To our knowledge, this is the first model of oligopolistic competition with both customer capital and productivity as dynamic state variables. This enables us to study the dynamic misallocation between productivity and customer capital due to strategic interactions.

In classic models of competition and innovation, rents are lowest when firms are neck and neck, and investment concentrates there (e.g., Aghion et al., 2005). Competition with variable markups changes this structure: when markups rise with market share, the return to widening a gap is largest for the firm already ahead, so the marketing incentive peaks when firms already have an advantage and large investments go to defending existing leadership. The strategic feedback raises the persistence of customer capital above what it would be if it only complemented firm productivity. Firms with identical productivity can therefore develop persistent size dispersion, and marketing alone can sustain concentration. We term this a *legacy effect*: past marketing raises current marketing with productivity held fixed, so firms can hold high market shares by defending accumulated customer capital, slowing leadership churn.

With demand imperfectly co-moving with productivity, concentration can be good or bad: productivity-driven concentration raises measured TFP and demand-driven concentration lowers it, at identical shares and markups in the two cases. The sign of the aggregate relationship therefore depends on whether demand sits on the productive firms, and we take it to the data.

We study the quantitative implications of the model for concentration and aggregate

productivity. Reallocating customer capital to equate its revenue product would raise output by 52%, following [Hsieh and Klenow \(2009\)](#) with customer capital in place of capital; the dynamics of accumulation sustain the dispersion behind it. Persistence alone cannot say whether this dispersion reflects efficient sorting of demand toward productive firms or distortionary entrenchment of incumbents, and we thus use the model to guide this analysis.³

The calibrated model splits this misallocation into its dynamic drivers: 44% reflects adjustment costs, the loss from investment that tracks productivity only with a lag, and 11% reflects the legacy effect, firms defending accumulated demand that the efficient allocation would reassign. The legacy effect on its own thus carries an efficiency cost of about 5%: this is what a planner recovers by correcting the strategic incentive while still paying the same cost of adjusting demand that firms do. The remaining 45% is a residual outside the model’s dynamic channels. The three shares are of the 41.9 log points that the 52% gain represents, so the legacy effect is 4.7 log points.

In the quantified framework, marketing lowers welfare on net. It aligns demand with productive firms, but the concentration and markups it sustains, together with the labor it absorbs, outweigh that gain: shutting marketing down lowers frontier productivity by 4.9% while welfare rises by 5.2%. We then study size-based policies. Because private returns to marketing rise with the accumulated stock while social returns fall with it, instruments that reward size, output subsidies among them ([Edmond et al., 2023](#)), can push marketing in the wrong direction. A small output subsidy still increases welfare; a large one entrenches incumbents, which continue to invest in building demand even with low productivity.

The remainder of this section reviews the literature, while the rest of the paper is structured as follows. Section 2 documents the marketing facts using firm-level data. Section 3 introduces the theoretical model with oligopolistic firms with heterogeneous productivity and customer capital. Section 4 measures the misallocation between customer capital and productivity and calibrates the dynamic game to the estimated dynamics. Section 5 derives the planner’s solution and uses counterfactuals to study how the quantified model changes our understanding of size-based policies, marketing, and

³NielsenIQ scanner data corroborate the construct: the [Hottman et al. \(2016\)](#) residual-demand measure behaves like the marketing-based stock, and its entry and exit flows give an independent check on the model’s decay and marketing-cost parameters (Appendix C).

antitrust; Appendix C corroborates the customer-capital construct in NielsenIQ scanner data.

Related Literature. A growing literature studies how firms build their customer base, treating demand as a slow-moving state that firms accumulate through investment (Dinlersoz and Yorukoglu, 2012; Drozd and Nosal, 2012; Gourio and Rudanko, 2014; Ignaszak and Sedláček, 2022; He et al., 2024; Chiavari, 2026). Demand persists through consumer inertia (Bornstein, 2019), through brand preferences that hold across decades and locations (Bronnenberg et al., 2009, 2012), and through relationship capital in buyer–seller settings (Heise, 2024); young plants, entrants, and exporters grow by accumulating demand (Foster et al., 2008, 2016; Fitzgerald et al., 2024; Argente et al., 2025), and customers, rather than sales per customer, account for the bulk of the variation in merchant sales in transaction data (Einav et al., 2021; Afrouzi et al., 2023); and in product-level data residual demand carries the largest share of the variation in firm size (Hottman et al., 2016; Eslava et al., 2024), with product creation central to sustaining it (Argente et al., 2024).

Customer capital is an essential intangible asset of firms (Haskel and Westlake, 2017; Crouzet and Eberly, 2019; Bronnenberg et al., 2022; Pearce and Wu, 2024). Kaplan and Zoch (2020) note the rising role of labor in activities related to expanding firm share, such as marketing. There is a rising literature studying this (Cavenaile and Roldan-Blanco, 2021; Afrouzi et al., 2023; Cavenaile et al., 2023; Bornstein and Peter, 2025; Cavenaile et al., 2025; Hubmer and Nord, 2025), yet the strategic role of marketing and path dependency is underemphasized. Afrouzi et al. (2023) also connect customer acquisition to market power and misallocation; we differ in this paper by focusing on an oligopolistic setting, where firms defend the stock they have accumulated, leading to legacy effects. The closest antecedent in strategic competition for customers is Cavenaile et al. (2025), where advertising shifts demand across oligopolists and firms strategically invest in marketing. The key difference is that the investments are period-by-period and do not treat customer capital as a state variable, whereas we focus on customer capital as a dynamic state variable, motivated by our initial facts presented in this section.

Our approach builds on the firm-dynamics tradition (Jovanovic, 1982; Hopenhayn, 1992; Klette and Kortum, 2004; Gabaix, 2011; Baqaee and Farhi, 2019), and on the misallocation of factors across firms (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009;

David and Venkateswaran, 2019). How firms invest in innovation, and how market structure shapes that investment, is the subject of a second branch (Aghion et al., 2005; Peters, 2020; Liu et al., 2022; Akcigit and Ates, 2023), built mostly to answer growth questions. We relate these two literatures through studying investment in an oligopolistic setting. We turn the investment machinery common for R&D in the growth literature into a misallocation question when investment is not inherently productivity-enhancing and firms have two dimensions of heterogeneity. Our framework also stays tractable at empirically relevant firm counts, so concentration is a quantitative equilibrium object rather than a feature fixed by the market structure.

A third literature studies rising market power and its efficiency implications (Atkeson and Burstein, 2008; De Loecker and Eeckhout, 2018; De Loecker et al., 2020; Olmstead-Rumsey, 2022; De Ridder, 2024; Mongey and Waugh, 2025); the rise of superstar firms and the labor share decline (Autor et al., 2020; Kehrig and Vincent, 2021) are consistent with a significant role for demand, in line with our facts. Covarrubias et al. (2020) discuss the nature of good versus bad concentration across U.S. industries over thirty years; in our model the two kinds coexist state by state, and the sign of the concentration–productivity relationship depends on whether demand sits on the productive firms. Firm size has more than one source (Chan et al., 2024). When size reflects productivity alone, correcting the static markup distortion and raising efficiency go hand in hand (Edmond et al., 2023); with a second, demand-driven source of size the two separate. We extend the insight of Atkeson and Burstein (2008) that variable markups generate misallocation into a dynamic setting, where a size-based instrument faces a trade-off between compressing markups and degrading the dynamic sorting of customers toward productive firms: larger is not necessarily better, and this message is central to our discussion and quantitative results.

2 Measurement and Facts of Customer Capital

We construct customer capital from firm-level marketing data and document three facts: it tracks firm size more closely than productivity does, it is more persistent than productivity and its investment is path dependent, and markets are granular. These facts motivate the dynamic model of Section 3; Section 4 then measures the misallocation between size and productivity that they imply.

2.1 Data and Measurement

We draw on Capital IQ matched to Compustat for firm-level marketing expenditures, revenue, and employment of publicly traded U.S. firms, with sales and marketing expenditure classified as in He et al. (2024). Capital IQ reports marketing from 1997; we accumulate the stock from that year and take 2007–2022 as the analysis panel to allow for accumulation of customer capital stock. We focus on two data samples. First, we focus on firms that have available direct sales and marketing data from CapitalIQ and He et al. (2024). The panel has 30,816 firm-years across roughly 4,100 firms with directly reported marketing. Second, we impute marketing for firms that report SG&A but not marketing, at the industry $\times$ year median marketing/SG&A ratio, which expands the panel to 56,349 firm-years across roughly 6,500 firms.⁴ All dollar variables are deflated to 2017 real dollars.

We construct a stock of customer capital via the perpetual inventory method:

$$b_{it} = (1 - \delta) b_{i,t-1} + m_{it}, \quad (1)$$

where m_{it} is real marketing expenditure and $\delta = 0.15$ is the annual depreciation rate.⁵ The stock capitalizes what the firm spends marketing dollars on. Demand also may accrue through channels the firm does not purchase, such as word of mouth. Section 4.2 treats this unpurchased accumulation explicitly and identifies its variance from the dynamics of revenue independent of productivity. In this section, we measure labor productivity as log real revenue per worker.⁶ Table 1 reports the moments behind the three facts for both samples.

Fact 1: customer capital tracks firm size. Customer capital correlates with log revenue at 0.73–0.83 across the two samples, while log revenue per worker correlates with log

⁴For firms without observed marketing, we compute the NAICS-3 $\times$ year median ratio of marketing to SG&A among firms reporting both, then multiply by the non-reporting firm’s SG&A. All the main results are robust to this classification choice.

⁵We follow He et al. (2024) for the construction of marketing expenditures and use the depreciation rate from purchase price allocation useful-life estimates. Our results are robust to alternative depreciation rates $\delta \in \{0.05, 0.10, 0.20, 0.25, 0.30\}$. Appendix C adds a structural cross-check in NielsenIQ scanner data, decomposing market shares into demand, productivity, and markups following Hottman et al. (2016) to recover a residual-demand measure of customer capital.

⁶The misallocation and calibration exercises of Section 4 instead use a revenue-productivity residual that nets customer capital out of revenue. Both measures show similar persistence and correlation with firm size.

TABLE 1: CUSTOMER CAPITAL, PRODUCTIVITY, AND EMPLOYMENT

	Customer Capital (b)	Productivity (a)	Employment
<i>Panel A: Marketing Sample (N = 30,816 firm-years)</i>			
Corr(log revenue, ·)	0.73	0.53	0.93
AR(1) coefficient	0.98	0.86	0.99
Correlation (x_t, x_{t-1})	0.99	0.88	0.99
AR(1), firm FE	0.81	0.49	0.79
<i>Panel B: Imputed Sample (N = 56,349 firm-years)</i>			
Corr(log revenue, ·)	0.83	0.50	0.92
AR(1) coefficient	0.98	0.88	0.99
Correlation (x_t, x_{t-1})	0.99	0.90	0.98
AR(1), firm FE	0.89	0.50	0.78

Notes: Correlations and AR(1) coefficients computed within NAICS-3 $\times$ year cells. Marketing sample: customer capital is log b_{it} from capitalized marketing expenditure ($\delta = 0.15$). Imputed sample adds firms where marketing is imputed from the NAICS-3 $\times$ year median marketing/SG&A ratio. Productivity is log revenue per worker; employment is log headcount. AR(1) from OLS with robust standard errors. Firm FE: within-firm AR(1) with year dummies. Sample: publicly traded U.S. firms, 2007–2022.

revenue at 0.50–0.53. This motivates the role of demand as a central state variable and will be revisited in the misallocation accounting of Section 4.

Fact 2: customer capital is persistent and its investment is path dependent. The industry-demeaned AR(1) of the stock is 0.98, against 0.86 for productivity (with firm fixed effects, the difference is stronger: 0.81 versus 0.49). This high persistence suggests demand investment continues to have important effects over long horizons and suggests it is an important state variable. Further, the investment itself is path dependent: marketing growth is more positively autocorrelated (Appendix B.8), where physical capital investment growth is negatively autocorrelated (David and Venkateswaran, 2019). This motivates the legacy effect we study theoretically and quantitatively: firms’ past investments in demand lead to current investments in demand.

Fact 3: markets are granular. Competition for customer capital takes place among a handful of large firms. The sales-weighted inverse Herfindahl index gives 4–7 major competitors across market definitions, roughly the concentration of consumer packaged goods, where the market contains every barcode (Pearce and Wu, 2024). Firms are therefore large enough to internalize their effect on the market. The model builds on the

potential for this strategic effect.

All of these facts speak to important elements of the model. The importance of customer capital for firm size in both the level and persistence indicates customer capital's natural role as a state variable for firms. Given that customer capital investment is persistent and path dependent, its accumulation is a relevant force among a handful of large firms. Section 3 builds an oligopoly economy consistent with these facts. Firms invest in customer capital under two incentives, a *productivity incentive* that sorts demand toward productive firms and a *size incentive* through which variable markups reward large firms regardless of productivity, and Section 4 measures how much misallocation between size and productivity the two incentives generate.

3 Model

We study the dynamics emerging from these facts by developing a model in which oligopolistic firms hold two stocks: productivity, which evolves exogenously, and customer capital, which they accumulate through costly marketing. Granular firms here internalize that a larger market share carries a higher markup, so marketing responds to size itself and demand acquires its own dynamics. Our contribution is twofold: the model focuses on the drivers of demand investment in the strategic environment, and a new computational method solves the resulting dynamic game globally at empirically relevant firm counts.

Section 3.1 lays out the environment, and Section 3.2 reduces the aggregate equilibrium to self-contained industry blocks. Section 3.3 characterizes the static pricing equilibrium within an industry and the paper's central asymmetry: profits depend only on the sum of productivity and customer capital, while efficiency depends on the split. Section 3.4 defines the dynamic marketing equilibrium; Section 3.5 works out the two-firm case, in which the economics is analytical, and Section 3.6 shows how we compute the equilibrium at general firm counts. Section 3.7 derives the welfare accounting used in the quantitative analysis. Proofs and derivations for this section are collected in Appendix A.

3.1 Environment

Time is discrete. The economy has a unit continuum of industries. Within each industry, there are I firms. A representative household consumes from products in these industries.

Household. We denote the aggregate consumption of the household as $\mathcal{C}_t$ and its labor supply as $\mathcal{L}_t$. The representative household optimizes the following problem:

$$\max_{c_{ikt}, \mathcal{L}_t} \sum_{t=0}^{\infty} \beta^t (\ln \mathcal{C}_t - \mathcal{L}_t), \quad (2)$$

subject to the budget constraint $\int_0^1 \sum_{i=1}^I p_{ikt} c_{ikt} dk = \mathcal{L}_t + \Pi_t$, where p_{ikt} and c_{ikt} are the price and consumption of products from firm i within industry k . We assume the total profit of the economy, Π_t , is also rebated to the household. Throughout the paper, we normalize the wage for labor to 1. The consumption at firm level is aggregated via a nested constant-elasticity-of-substitution (CES) system. More precisely, the aggregate consumption is a Cobb–Douglas aggregation from consumption in the continuum of industries

$$\ln \mathcal{C}_t = \int_0^1 \ln C_{kt} dk, \quad (3)$$

and the industry-level consumption C_{kt} follows a CES aggregation

$$C_{kt} = \left[\sum_{i=1}^I \left(\frac{b_{ikt}^\nu}{\sum_j b_{jkt}^\nu} \right)^{1/\sigma} c_{ikt}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \quad (4)$$

The aggregation of firm-level consumption among firms follows an otherwise standard CES system with $\sigma > 1$ being the elasticity. Our model differs by the forming of consumption weights in the demand system. More precisely, we assume the weight for consumption of firm i is proportional to its measured capitalized marketing expenditure, with $\nu > 0$ being the customer capital elasticity. Customer capital captures the household's prior tendency to buy from a firm before observing prices, the product's image or its shelf presence.⁷ Because customer capital enters only through the normalized weight, customer capital is purely allocative and does not bring level-effects on

⁷Appendix A.6 microfound this demand system from rational inattention in consumption choices.

consumption.

Firms. Firm i produces with labor at a constant elasticity $\omega \in (0, 1]$,

$$c_{it} = a_{it}^{1/(\sigma-1)} \ell_{it}^\omega, \quad (5)$$

where a_{it} is its productivity scaled into market-share units.⁸ Decreasing returns arise from a factor fixed within the period, which collapses into (5) and leaves the demand side untouched. Productivity follows an AR(1) process

$$\ln a_{i,t+1} = \rho_a \ln a_{it} + \varepsilon_{i,t+1}^a, \quad (6)$$

with $\rho_a \in (0, 1)$ the persistence of log productivity and independent Gaussian innovations $\varepsilon_{i,t+1}^a \sim N(0, \nu_a^2)$.

The dynamic decision of firms is the choice of marketing expenditure. More precisely, we assume the firms' customer capital follows the law of motion

$$b_{ikt} = [(1 - \delta)b_{ik,t-1} + m_{ikt}] \varepsilon_{ikt}^b. \quad (7)$$

m_{ikt} is firms' marketing expenditure, the same data object we measured. ε_{ikt}^b represents the unmeasured demand shocks that are not directly affected by the marketing efforts. We assume the labor cost of marketing is given by

$$K_i(m_{ikt}, \mathbf{b}_{k,t-1}) = m_{ikt} + \frac{\kappa}{2} \left(\frac{b_{ik,t-1}}{\sum_j b_{jk,t-1}} \right)^\theta \left(\frac{m_{ikt}}{b_{ik,t-1}} \right)^2. \quad (8)$$

There are two parts to the cost of marketing. First, the direct labor expenditure m_{ikt} as we measure in the data; second, the convex cost that increases in the growth rate of customer capital $\frac{m_{ikt}}{b_{ik,t-1}}$. Additionally, we allow the marketing cost to depend on the current demand weight of firms through a flexible parameter θ . θ measures the cost disadvantage of large firms in marketing: when θ is more positive, a larger firm (according to demand-side heterogeneity) pays a larger cost in growing its customer capital.

⁸The scaling makes a one percent productivity gap map into a one percent market-share gap, holding customer capital fixed; the quantitative analysis treats the scaling explicitly.

Granularity. Firms are large relative to their market: their shocks move market aggregates, and they internalize their impact on the price index, competing in quantities as in Cournot.

3.2 From Aggregate to Industry Equilibrium

We set up the household problem so that the industry-level dynamics are independently analyzed without referring to aggregates. In this section, we discuss these results and leave the formal analysis of the household problem to Appendix A.8.

First, the linear disutility of labor supply and log utility in consumption means the real total expenditure of the economy is always 1. The Cobb–Douglas aggregation across the measure 1 of industries also means the total expenditure in each industry stays 1. This leads to the CES demand curve within industry k as

$$c_{ikt} = \frac{b_{ikt}^v}{\sum_j b_{jkt}^v} \left(\frac{p_{ikt}}{P_{kt}} \right)^{-\sigma} \frac{1}{P_{kt}}, \quad (9)$$

where $P_{kt} = \left(\sum_i \frac{b_{ikt}^v}{\sum_j b_{jkt}^v} p_{ikt}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$ is the industry-level price index. Correspondingly, the consumption within industry is $C_{kt} = 1/P_{kt}$.

Second, the aggregate labor market clearing requires

$$\mathcal{L}_t = \int_0^1 (L_{kt}^{\text{prod}} + L_{kt}^{\text{mkt}}) dk, \quad (10)$$

with the two industry labor pools defined in Section 3.7. Household optimality reduces to three conditions, derived in Appendix A.8. Linear disutility equates the marginal value of wealth to the wage, so with the wage as numeraire the marginal value of wealth is one and labor supply is perfectly elastic; the labor market absorbs any aggregate labor demand (10) without moving relative prices. Log aggregation across industries makes expenditure shares constant across k , and with the marginal value of wealth at one the common level is pinned,

$$E_{kt} \equiv \sum_{i=1}^I p_{ikt} c_{ikt} = 1 \quad \text{for every } k \text{ and } t, \quad (11)$$

so each industry's revenue is invariant to every aggregate and to every other industry's state. The portfolio condition prices firm profits at the household's discount factor β .

Together these conditions make the aggregate equilibrium block-recursive. Industry expenditure, the wage, and the discount rate are all constants, so the equilibrium of the aggregate economy is the equilibrium of a single industry, solved on its own state $(\mathbf{a}_k, \mathbf{b}_k)$ and replicated over the continuum. Sections 3.3 through 3.6 characterize the equilibrium of one industry, dropping the index k . Welfare is the one remaining aggregate object; it aggregates linearly in logs across the continuum, and Section 3.7 establishes that the aggregate flow is the ergodic mean of the per-industry flow even though individual industries never settle at a steady state.

3.3 Static Equilibrium

Within a period, given the industry state $(\mathbf{a}, \mathbf{b})$, quantity competition delivers markup pricing over marginal cost, $\mu_{it} = \frac{\sigma}{\sigma-1} \frac{1}{1-s_{it}}$, since the perceived elasticity is a property of residual demand and the share rather than of costs. Marginal cost rises with own output, so the price carries a scale term as well, $p_{it} = s_{it}^{1-\omega} \tilde{\mu}_{it}^\omega a_{it}^{-1/(\sigma-1)}$, and large firms price higher for a second reason beyond market power.

Writing $z_{it} \equiv \ln a_{it} + \nu \ln b_{it}$ for the *composite* and

$$\tilde{\sigma} \equiv \frac{\sigma}{\sigma - \omega(\sigma - 1)}, \quad \hat{z}_{it} \equiv \frac{\tilde{\sigma}}{\sigma} z_{it}, \quad (12)$$

shares solve the fixed point

$$s_{it} = \frac{e^{\hat{z}_{it}} (1 - s_{it})^{\tilde{\sigma}-1}}{\sum_{j=1}^I e^{\hat{z}_{jt}} (1 - s_{jt})^{\tilde{\sigma}-1}}, \quad (13)$$

profits are a convex function of the own share alone,

$$\pi_{it} = \frac{1}{\tilde{\sigma}} s_{it} + \frac{\tilde{\sigma} - 1}{\tilde{\sigma}} s_{it}^2, \quad (14)$$

and the average-cost markup is $\tilde{\mu}_{it} = \frac{\tilde{\sigma}}{\tilde{\sigma}-1} \frac{1}{1-s_{it}}$, so the production labor pool is the reciprocal of the share-weighted harmonic mean of those markups, $L^{\text{prod}} = 1/\tilde{\mu}$. Returns to scale act on the cost side and are common to both stocks, so the composite summarizes the state, and $\tilde{\sigma}$ sets the rate at which a unit of it moves the share. The static equilibrium

depends on σ and ω only through $\tilde{\sigma}$, so the two are not separately identified by static targets.

Profits depend on the state only through the composite z_{it} : for the firm, a unit of productivity and a unit of customer capital are perfect substitutes. Real consumption does not share this symmetry. With labor allocated by the pricing stage, $\ell_i = s_i / \tilde{\mu}_i$,

$$C^{\frac{\sigma-1}{\sigma}} = \sum_i \left(\frac{b_i^v}{\sum_k b_k^v} \right)^{1/\sigma} a_i^{1/\sigma} \left(\frac{s_i}{\tilde{\mu}_i} \right)^{\frac{\omega(\sigma-1)}{\sigma}}, \quad (15)$$

so productivity raises output directly while customer capital only reweights it across firms; by the normalization in (4), scaling every firm's stock by a common factor leaves output unchanged, so only the distribution of demand matters. Appendix A.1 derives the fixed point, marginal profits, and the output expressions behind (13)–(15).

3.4 Dynamic Equilibrium

We focus on Markov perfect equilibria: each firm's strategy depends on the payoff-relevant state alone. Because all firms are granular in our model, the relevant state for their dynamic decision is the entire vector of productivity $\mathbf{a} = (a_{it})_{i=1}^I$ and the vector of customer capital $\mathbf{b} = (b_{it})_{i=1}^I$. We denote the discounted value of firm i given $(\mathbf{a}, \mathbf{b})$ as $V_i(\mathbf{a}, \mathbf{b})$ and its marketing expenditure as $m_i(\mathbf{a}, \mathbf{b})$. The dynamic problem of firm i is

$$V_i(\mathbf{a}, \mathbf{b}) = \max_{m_i \geq 0} \left\{ \pi_i(\mathbf{z}) - m_i - \frac{\kappa}{2} \left(\frac{b_i}{\sum_j b_j} \right)^\theta \left(\frac{m_i}{b_i} \right)^2 + \beta \mathbb{E}[V_i(\mathbf{a}', \mathbf{b}')] \right\}, \quad (16)$$

subject to the laws of motion (6) and (7) for all firms, including the feedback of rivals' strategies into rivals' states.

Definition 1 A marketing equilibrium is a profile of strategies $\{m_i(\mathbf{a}, \mathbf{b})\}_{i=1}^I$ such that each m_i attains the maximum in (16) given the others.

The optimal choice of marketing expenditure satisfies the first-order condition. We express this condition using the investment rate of customer capital due to marketing $\eta_i = \frac{m_i}{b_i}$:

$$\eta_i(\mathbf{a}, \mathbf{b}) = \max \left\{ \frac{b_i}{\kappa} \left(\frac{b_i}{\sum_j b_j} \right)^{-\theta} \left(\beta \mathbb{E} \left[\frac{\partial V_i}{\partial b'_i}(\mathbf{a}', \mathbf{b}') \varepsilon_i^b \right] - 1 \right), 0 \right\} \quad (17)$$

Inside equation (17), firms invest in customer capital only when the marginal value of customer capital $E \left[\frac{\partial V_i}{\partial b'_i}(\mathbf{a}', \mathbf{b}') \right]$ is large enough; otherwise, they do not invest and let the existing customer capital depreciate.

In order to compute the marginal value of customer capital, we need to further derive the marginal value from the Bellman equation:

$$\frac{\partial V_i}{\partial b_i} = \frac{\partial \pi_i}{\partial b_i} - \frac{\partial K_i}{\partial b_i} + (1 - \delta) \beta \mathbb{E} \left[\frac{\partial V_i}{\partial b'_i}(\mathbf{a}', \mathbf{b}') \varepsilon_i^b \right] + \sum_{j \neq i} \frac{\partial m_j}{\partial b_i} \beta \mathbb{E} \left[\frac{\partial V_i}{\partial b'_j}(\mathbf{a}', \mathbf{b}') \varepsilon_j^b \right] \quad (18)$$

The last term in the envelope condition is the strategic feedback: firm i 's state moves rivals' future marketing, and firm i internalizes that response. This term is absent under atomistic competition.

Computing this equilibrium is difficult on two counts. The strategic feedback makes each firm's policy depend on the derivatives of rivals' policies, a fixed point in a space of functions; and the state is the full cross-section $(\mathbf{a}, \mathbf{b})$, so grid-based methods face a state space that grows exponentially in I . We take the two routes in turn. The next section shows that the inclusion of Gaussian investment shocks allows us to capture this strategic interaction without computing the cross derivatives.

Stein representation. With Gaussian innovations in logs, $\ln \varepsilon_i^b \sim N(0, \nu_b^2)$, the expected marginal value of customer capital can be calculated as the covariance between the shock and the level of value:

$$\eta_i(\mathbf{a}, \mathbf{b}) = \max \left\{ \frac{b_i}{\kappa} \left(\frac{b_i}{\sum_j b_j} \right)^{-\theta} \left(\frac{\beta}{\nu_b^2} \frac{\mathbb{E} [\ln \varepsilon_i^b V_i(\mathbf{a}', \mathbf{b}')] }{(1 - \delta)b_i + m_i} - 1 \right), 0 \right\} \quad (19)$$

Appendix A.2 gives the derivation. The advantage of equation (19) is that it does not require us to compute the strategic interactions per se. Instead, together with the Bellman equation in (16), the dynamic equilibrium becomes a recursion in terms of (V_i, η_i) .

3.5 Two-Firm Case

With two firms the game's economics is analytical. If exogenous persistence coincides ($\rho_a = \rho_b \equiv \rho$), profits depend on the state only through the composite gap $\zeta \equiv z_1 - z_2$, and the equilibrium reduces to one dimension. The two-firm case establishes, in closed

form, the three results the quantitative analysis scales up: the marketing incentive peaks at a strict lead, so competition rewards incumbency; the strategic feedback raises the persistence of market positions above its technological rate; and concentration is good or bad for measured productivity depending on whether it is productivity-driven or demand-driven. Throughout, statements are exact unless labeled numerical; numerical statements use a fitted equilibrium (computed by the method of Section 3.6) at illustrative parameters. Appendix A.3 collects the proofs, the parameters, and every magnitude.

The share fixed point (13) solves in closed form at $I = 2$: the leader's share is logistic in the gap,

$$s_1(\zeta) = \frac{1}{1 + e^{-\zeta/\sigma}}, \quad (20)$$

with slope $s'(\zeta) = s(1-s)/\sigma$. The flow incentive to widen the gap combines how the share responds to the gap with how profit responds to the share, $h(\zeta) \equiv d\pi_1/d\zeta = s'(\zeta) \pi'(s)$.

Proposition 1 (Incentive peaks at a strict lead) *With profits (14), the flow incentive $h(\zeta) = s(1-s) [1 + 2(\tilde{\sigma} - 1)s] / (\sigma\tilde{\sigma})$ is maximized at the unique share*

$$s^* = \frac{(2\tilde{\sigma} - 3) + \sqrt{(2\tilde{\sigma} - 3)^2 + 6(\tilde{\sigma} - 1)}}{6(\tilde{\sigma} - 1)} \in \left(\frac{1}{2}, 1\right),$$

with $s^ \rightarrow \frac{1}{2}$ as $\tilde{\sigma} \rightarrow 1$. If profit were linear in the share (constant markups), $h \propto s(1-s)$ would peak exactly at the tie.*

Locally the result is a statement about profit curvature alone: since $s''(0) = 0$, the slope of the incentive at the tie is $h'(0) = s'(0)^2 \pi''(\bar{s})$, so the sign of π'' classifies the peak. Convex profit, the Cournot case where markups rise with share, puts the peak on the leader's side; concave profit would put it on the laggard's side and generate catch-up instead. At conventional elasticities the peak sits well inside the leader's territory, and the fitted equilibrium policy, which aggregates discounted future flow incentives, peaks at a still larger lead (numerical). In the step-by-step innovation models of Aghion et al. (2005), rents are lowest when firms are neck and neck, so the incentive peaks at the tie and firms invest to escape competition; here interior duopoly rents survive the tie, and the same escape-competition logic pushes the peak to a strict lead, so investment defends leadership rather than contesting it.

Endogenous persistence. The gap follows the closed loop $\zeta' = \rho\zeta + \Delta\eta(\zeta) + \varepsilon_\zeta$, where $\Delta\eta \equiv \eta_1 - \eta_2$ is the equilibrium marketing difference and $\varepsilon_\zeta \sim N(0, v_\zeta^2)$ with $v_\zeta^2 = 2(v_a^2 + v_b^2)$. The slope $\gamma \equiv \Delta\eta'(0)$ measures the strategic feedback: whether a widening gap induces marketing that widens it further.

Proposition 2 (Feedback is positive) *To leading order in β , the feedback is the shock-weighted asymmetry of the flow incentive,*

$$\gamma = \frac{\beta\rho}{\kappa v_\zeta^2} \mathbb{E} \left[\varepsilon (h(\varepsilon) - h(-\varepsilon)) \right] + O(\beta^2),$$

where

$$h(\varepsilon) - h(-\varepsilon) = \frac{2(\tilde{\sigma} - 1)}{\sigma\tilde{\sigma}} s(1-s)(2s-1) \geq 0,$$

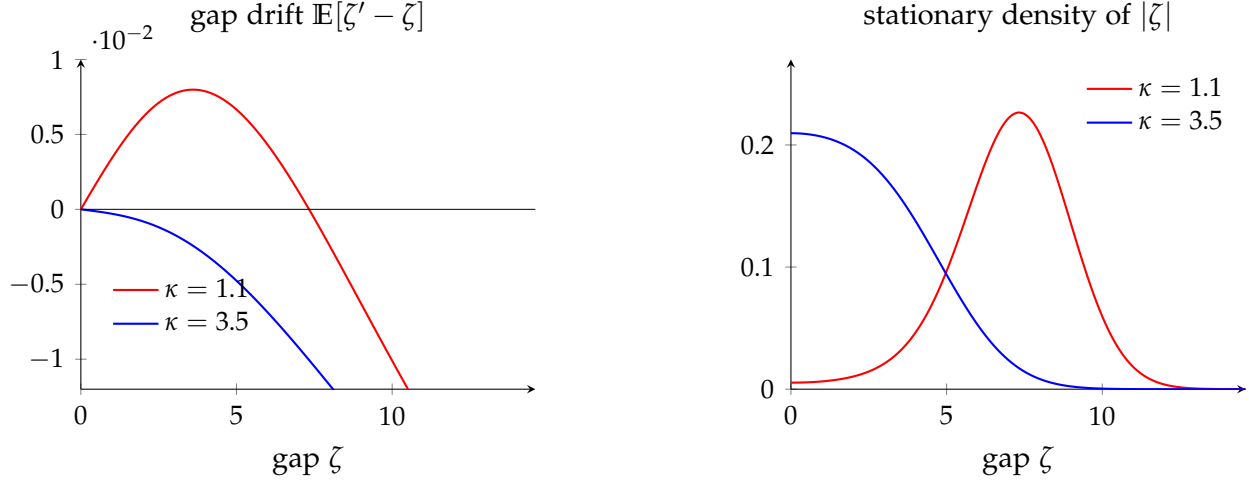
with strict inequality for $\tilde{\sigma} > 1$, and $\gamma = 0$ exactly when profit is linear in the share.

Positive feedback raises the persistence of market positions: the closed-loop gap behaves like an AR(1) with coefficient $\rho + \gamma$. At the illustrative parameters the fitted feedback is positive and modest, amplifying the stationary variance of the gap (numerical). This wedge between measured and technological persistence is the *legacy effect*; the calibration in Section 4.2 identifies it in the data. The feedback also admits a threshold: if γ exceeds the mean reversion $1 - \rho$, the symmetric state turns unstable and the stationary distribution of the gap peaks at a strict lead; marketing alone then sustains concentration. Figure 1 displays the threshold as a comparative static in the marketing cost: at a high marketing cost the gap drift is negative everywhere and the stationary distribution is unimodal at the tie, while at a low marketing cost the drift turns positive near the tie and the distribution peaks when a firm is leading, so concentration can arise without a productivity gap.

Internalization is essential to the sign. An atomistic firm facing a variable-elasticity (Kimball) demand has, by the envelope theorem, $d^2 \ln \pi / dz^2$ proportional to the slope of its demand elasticity, so its proportional return to the state *declines* in size exactly when markups rise with size: the atomistic analogue of γ is nonpositive. Amplification requires firms large enough to perceive the wedge between the margin and the physical elasticity; quantity competition among granular firms provides that scale.

Section 3.7 defines three productivity indices: the equilibrium index Z^{eq} , output per unit of production labor; the frontier Z^* , the index with labor reallocated efficiently at

FIGURE 1: TWO-FIRMS



Notes: Closed loop of the composite gap under the leading-order policy of Proposition 2, at $\sigma = 4$, $\rho = 0.998$, $\beta = 0.5$, $\nu_a = 0.1$, $\nu_b = 0$. Left: expected one-period change of the gap. Right: implied stationary density of the absolute gap, unimodal at the tie for $\kappa = 3.5$ and peaked at a strict lead for $\kappa = 1.1$. Source: model computations.

the same state; and their ratio, the markup wedge $\mathcal{M}$. In the gap coordinates $\alpha \equiv \ln a_1 - \ln a_2$, $\beta_b \equiv \nu \ln b_1 - \nu \ln b_2$, $\zeta = \alpha + \beta_b$, they take closed forms, stated here at $\omega = 1$ (where $\tilde{\sigma} = \sigma$) and carried to general ω in Appendix A.4:

$$\begin{aligned}
 Z^* &= \left[\frac{\cosh(\zeta/2)}{\cosh(\beta_b/2)} \right]^{\frac{1}{\sigma-1}}, \\
 \mathcal{M}(\zeta) &= \frac{[\cosh(\zeta/2)]^{\frac{1}{\sigma-1}}}{\left[\cosh\left(\frac{\zeta}{2\sigma}\right) \right]^{\frac{\sigma}{\sigma-1}}}, \\
 \ln Z^{\text{eq}} &= \ln Z^* - \ln \mathcal{M}.
 \end{aligned} \tag{21}$$

The markup wedge $\mathcal{M}$ depends on the composite alone, so it is invariant to whether a gap comes from productivity or from demand. All source dependence sits in the frontier Z^* , and the equilibrium index splits accordingly into an alignment term and a markup term. Underlying the closed forms is a property of Cournot pricing.

Proposition 3 (Labor invariance) *At $I = 2$ the two firms hire identical production labor at every state: $\ell_i = \frac{\tilde{\sigma}-1}{\tilde{\sigma}} s_i (1 - s_i)$ and $s_2 = 1 - s_1$ make $\ell_1 = \ell_2$ an identity.*

Both firms equate marginal revenue to the same wage, so the leader's entire composite advantage exits through its price. The equilibrium never reallocates production labor

toward the productive firm, so the markup wedge measures the productivity value of the reallocation that markups block.

Corollary 1 (Sign test) *Exactly,*

$$\left. \frac{\partial \ln Z^{\text{eq}}}{\partial \zeta} \right|_{\beta_b} = \frac{\tanh(\zeta/2\sigma)}{2(\sigma-1)} > 0,$$

while

$$\left. \frac{\partial \ln Z^{\text{eq}}}{\partial \beta_b} \right|_{\zeta} = -\frac{\tanh(\beta_b/2)}{2(\sigma-1)} < 0.$$

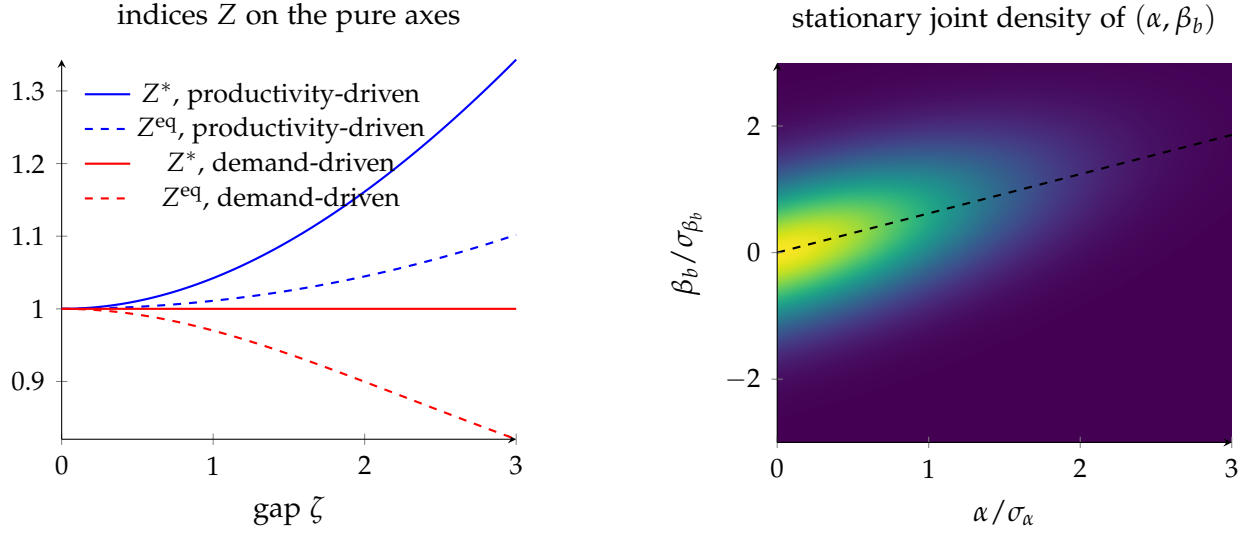
Concentration raises measured productivity when it is productivity-driven and lowers it when it is demand-driven, at identical shares and markups in the two cases. With customer capital equalized ($\beta_b = 0$), the frontier gain dominates the markup wedge at every gap, so productivity-driven concentration is always efficient; misallocation that hurts requires the demand dimension.

The concentration–productivity relationship has no fixed sign, and its slope inherits the sign of the alignment between demand and productivity. The left panel of Figure 2 traces the two indices along the pure axes: a productivity-driven gap raises both the frontier and the equilibrium index, while a demand-driven gap of the same size in shares and markups lowers the equilibrium index one for one with the markup wedge.

The matching between demand and productivity. Marketing is the endogenous variable that correlates productivity with customer capital. With $\Delta\eta$ linearized to $\gamma\zeta$, the stationary covariance of the productivity and demand gaps is proportional to γ : zero under constant markups, positive here. The linear approximation is accurate: the fitted policy is close to linear over the states the economy visits. The resulting correlation between the productivity and demand gaps is strongly positive (the right panel of Figure 2).

On average more productive firms have more demand, yet the ergodic distribution keeps recurring mass on misaligned states. Productivity moves with fresh innovations while the demand gap is a distributed lag of past marketing, so a productivity reversal leaves the old leader holding the accumulated customer capital. Exactly, the economy is less productive than its no-marketing counterpart ($Z^* < 1$) if and only if $\alpha(\alpha + 2\beta_b) < 0$, the entrenchment region. Customer capital tilts toward the less productive firm in a

FIGURE 2: TWO-FIRM PRODUCTIVITY INDICES



Notes: Left: exact closed forms (21). Right: ergodic joint density of the standardized productivity and demand gaps at the baseline parameters ($\sigma_\alpha = 0.29$, $\sigma_{\beta_b} = 0.04$, correlation 0.62), folded onto the half plane; dashed line: $\mathbb{E}[\beta_b | \alpha]$. Source: model computations.

sizable minority of states, and only rarely is the market leader itself the less productive firm (numerical).

On the ergodic distribution the equilibrium index gains almost nothing over the no-marketing benchmark, alignment net of entrenchment, while the marketing contest absorbs a first-order share of revenue (Appendix A.3 reports the magnitudes). The social cost of marketing at two firms is dissipation rather than misallocation, and the quantitative analysis shows the same ordering survives at the calibrated firm count.

3.6 Computation

Beyond two firms the equilibrium has no closed form. We solve it with a method that exploits the Gaussian innovations to handle the two obstacles, the strategic feedback and the growing state space. The method works on the stacked state $\mathbf{x} = (\mathbf{a}, \mathbf{b})$ with the laws of motion; when the persistence of each coincides the state reduces to the composite $\mathbf{z}$, as in the two-firm case. Derivations, the admissible basis families, and the accuracy audit against exact small- I solutions are in Appendix A.2.

Critic family. We approximate the value function by a sum of one-dimensional index functions of the stacked state, a linear-quadratic block plus M probit terms,

$$V_i(\mathbf{x}) \approx \vartheta_0 + \vartheta_1 (u_i \cdot \mathbf{x}) + \vartheta_2 (u_i \cdot \mathbf{x})^2 + \sum_{n=1}^M \vartheta_{n+2} \Phi\left(\frac{d_n \cdot \mathbf{x} - \bar{x}_n}{w_n}\right), \quad (22)$$

where an index $d_n \cdot \mathbf{x} = d_n^a \cdot \mathbf{a} + d_n^b \cdot \mathbf{b}$ loads the two stocks separately, and we restrict the directions in three ways. Directions are mean zero within each stock, because the equilibrium is invariant to a common shift in either stock and only relative positions matter. Profiles are *conjugate* to the Gaussian, so every expectation in (19) has a closed form and no numerical integration is needed. The approximant inherits the game's symmetry (invariance to permutations of the rivals) because each free direction is tied to the sum over its rival permutations. These restrictions do not reduce flexibility, since sums of index functions can approximate any continuous function. Because the method stores a coefficient vector rather than a function on a grid, it is unaffected by the dimension of the state.

Fitting as regression. With the actions held at the current iterate, the Bellman residual of (19) rearranges to a linear regression of current flow payoffs on *annuitized* basis functions, where each regressor is the basis function minus β times its expected next-period value. The algorithm iterates the regression and the action update to a joint fixed point on a simulated ergodic panel, adding probit terms only when the residual remains large. Because the one-step-ahead term is computed in closed form, the regressors contain no simulation noise and the associated bias is absent. We audit every fit four ways: the Bellman residual out of sample, the distance to an exact small- I solver, the second-order condition, and a stationarity certificate requiring the closed-loop persistence of every pairwise gap $z_i - z_j$ to lie below one. The fitted M measures how far the equilibrium departs from its linear-quadratic benchmark. The fitted policies match exact solutions at small firm counts to a few parts in ten thousand; Appendix A.2 reports the audit (Figure A1 and Table A1).

3.7 Aggregation and Welfare

Individual industries never reach a steady state. The aggregate economy nonetheless has one, because industries are ex-ante identical and hit by independent shocks: at any date the cross-section of industry states is the stationary distribution of the industry equilibrium, and the aggregates are deterministic constants. We denote the stationary density of the industry-level state as $g(\mathbf{a}, \mathbf{b})$. The aggregate welfare can thus be written as the mean value of real consumption and labor input. More precisely, the welfare of the household in a steady state is $\frac{1}{1-\beta} \mathcal{W}$, where $\mathcal{W} = \mathbb{E}_g [\log C(\mathbf{a}, \mathbf{b}) - L^{\text{prod}}(\mathbf{a}, \mathbf{b}) - L^{\text{mkt}}(\mathbf{a}, \mathbf{b})]$. In Appendix A.4, we provide the details of the following decomposition of the aggregate welfare.

Lemma 1 (Welfare decomposition) *In an aggregate steady state, the discounted utility of the representative household is $\frac{1}{1-\beta} \mathbb{E}_g [W(\mathbf{a}, \mathbf{b})]$, with*

$$W(\mathbf{a}, \mathbf{b}) = \underbrace{\ln Z^*}_{\text{Efficient Productivity}} - \underbrace{\ln \mathcal{M}}_{\text{Markup Dispersion}} - \underbrace{\Lambda}_{\text{Markup Level}} - \underbrace{L^{\text{mkt}}}_{\text{Marketing Cost}}, \quad (23)$$

where

$$\begin{aligned} Z^* &= \left[\frac{\left(\sum_i (a_i b_i^\nu)^{\tilde{\sigma}/\sigma} \right)^{\sigma/\tilde{\sigma}}}{\sum_k b_k^\nu} \right]^{\frac{1}{\tilde{\sigma}-1}}, & \mathcal{M} &= \left[\frac{\left(\sum_i s_i \tilde{\mu}_i^{\tilde{\sigma}-1} \right)^{\frac{1}{\tilde{\sigma}-1}}}{\bar{\mu}} \right]^\omega, \\ \Lambda &= \frac{1}{\bar{\mu}} + \omega \ln \bar{\mu}, & L^{\text{mkt}} &= \sum_i \left[m_i + \frac{\kappa}{2} \left(\frac{b_i}{\sum_j b_j} \right)^\theta \eta_i^2 \right], \end{aligned}$$

and

$$\tilde{\mu}_i \equiv \frac{\tilde{\sigma}}{\tilde{\sigma}-1} \frac{1}{1-s_i}, \quad \bar{\mu} \equiv \left(\sum_i s_i \tilde{\mu}_i^{-1} \right)^{-1}. \quad (24)$$

Z^* is the frontier: output when production labor is allocated efficiently at the given state. Customer capital enters only through the normalized weights, so Z^* is invariant to the scale of the stock and moves only with its distribution: it rises when demand is allocated to productive firms, and under decreasing returns it also falls as demand concentrates at a given productivity, since a concentrated weight crowds output into the steep part of the technology.

$\mathcal{M}$ is the dispersion cost of markups. The equilibrium allocates labor by $\ell_i = s_i / \tilde{\mu}_i$,

which distorts the efficient allocation by a power of the markup alone, $\ell_i \propto \ell_i^* \tilde{\mu}_i^{-\tilde{\sigma}}$; the output loss is the gap between the power mean and the harmonic mean of the same markup distribution. It equals one exactly when markups are equal, however concentrated the market.

Λ is the level cost of markups. The harmonic markup sets the labor level through $L^{\text{prod}} = 1/\bar{\mu}$, and $\Lambda = L^{\text{prod}} - \omega \ln L^{\text{prod}}$ is that labor's linear disutility net of its log consumption value; the identity $\omega \ln \bar{\mu} = -\omega \ln L^{\text{prod}}$ makes the netting exact. It is minimized at $\bar{\mu} = 1/\omega$, the markup whose price–average-cost gap pays the fixed factor's rent: under decreasing returns some markup is efficient, and movements in Λ price only the excess. High markups at the largest firms reduce their hiring, so the labor level falls with concentration.⁹ L^{mkt} is the marketing bill, the measured spend plus the convex adjustment cost.

The decomposition is exact and the channels are exhaustive: each term is a function of the state, so counterfactual tables difference ergodic means channel by channel and the columns sum to the welfare change. The split mirrors [Edmond et al. \(2023\)](#): Λ is their uniform-markup channel and $\ln \mathcal{M}$ their misallocation loss; the two reappear separately in the Hsieh–Klenow accounting of Section 4.1. A productivity index alone omits Λ and L^{mkt} .

The equilibrium has two externalities: markup dispersion misallocates production labor, and firms, which are paid on the composite, do not internalize the social value of allocating customer capital to productive firms. Section 5 derives the planner's solution and quantifies both.

4 Quantification

This section studies the quantified model that links static allocation to dynamic choices of firms. We calibrate the model to the Compustat and Capital IQ panel of Section 2 and quantify the misallocation between demand and productivity.¹⁰ We then use the model

⁹Production labor depends on shares alone: with expenditure fixed at one, a productivity improvement lowers prices and raises quantities one for one in revenue terms, so it changes output per worker but not employment. The property follows from unit expenditure, the log upper tier, and linear labor disutility.

¹⁰The NielsenIQ scanner data of Appendix C corroborate the customer-capital construct and the dynamics with an independent, structural measure of demand.

to study the positive relationship between concentration and productivity before turning to normative and policy implications in Section 5.

We identify the parameters in three blocks. The first block consists of externally calibrated parameters: the elasticity of substitution, the discount factor, and the number of firms per market. The second is the static block, which rests on the firm-level production function for revenue and requires no assumption about the dynamics. Section 4.1 estimates it and reports the gain from reallocating customer capital, following the accounting common in work on capital misallocation (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). The third is dynamic and asks about the endogenous drivers of the static allocation: adjustment costs, legacy effects, and shocks. Section 4.2 estimates it first as a reduced-form law of motion and then structurally, and splits the static gap along those lines. Section 4.3 then computes concentration and productivity inside the calibrated game. Table 3 collects the parameters.

4.1 Static Misallocation

We build on the literature that studies the misallocation implications of the gap between the size and productivity of firms (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). We apply that accounting to customer capital, connecting it to the productivity concept from the model. We first discuss the identification of the main static parameters of the model, and we then turn to the discussion of misallocation.

Parameter identification We focus on revenue instead of market share and employ industry by year fixed effects, so that $R_i = s_i$ as in the model. Firm revenue output is a function of three components,

$$\log R_i = \alpha_L \log \ell_i + \alpha_B \log b_i + \log \text{TFPR}_i. \quad (25)$$

This is the revenue production function in both the model and the data: b_i is the customer-capital stock of (1), α_L is the revenue elasticity of labor, $\alpha_B = \nu/\sigma$ is the revenue elasticity of the stock, the direct effect of the demand weight b_i^ν through inverse demand, and the residual, revenue productivity TFPR_i , collects what revenue contains beyond the two inputs: the markup, true productivity, and the customer capital the firm did not purchase.

Given that labor is chosen conditional on the stock of customer capital, the coefficient on labor will be biased upward. Since labor is the freely flowing factor, chosen within each period at the constant equilibrium wage w , we can write out its first-order condition, $w\ell_i = (\omega/\mu_i)s_i$, which makes labor proportional to revenue deflated by the markup: given the share, labor moves one for one with revenue. Substituting it out concentrates (25) into

$$\log R_i = \frac{\alpha_B}{1 - \alpha_L} \log b_i + \log \tilde{a}_i + \text{const}, \quad \log \tilde{a}_i \equiv \frac{\log \text{TFPR}_i}{1 - \alpha_L}, \quad (26)$$

a regression on the stock alone: collecting the $\log R_i$ terms divides every coefficient by $1 - \alpha_L$, because labor moves with revenue and amplifies whatever moves revenue. The slope is therefore the total elasticity, $\bar{\alpha}_B \equiv \alpha_B/(1 - \alpha_L) = d \log R_i / d \log b_i = 0.598$, the elasticity the reallocation experiment needs, since moving customer capital between firms moves their shares too, and the residual $\tilde{a}_i$, revenue productivity rescaled the same way, is the object every later step uses. Every estimation step in this section runs through (25): its coefficient delivers the elasticities here, and the dynamics of its residual deliver the volatilities of Section 4.2.

Substituting the labor condition into the technology (5) gives the share fixed point $s_i = \hat{a}_i b_i^{\bar{\alpha}_B} (1 - s_i)^{\tilde{\sigma}-1}$, the levels version of (13), where $\hat{a}_i \equiv a_i^{\tilde{\sigma}/\sigma}$ is productivity on the rescaled index of (12) and the $1 - s_i$ term is the Cournot markup entering negatively.

The markup therefore sits in the residual's interpretation, not the estimation: the model reads

$$\log \tilde{a}_i = \log \hat{a}_i + (\tilde{\sigma} - 1) \log(1 - s_i) = \frac{\tilde{\sigma}}{\sigma} \log a_i + (\tilde{\sigma} - 1) \log(1 - s_i), \quad (27)$$

so the estimated residual $\tilde{a}_i$ is the structural index $\hat{a}_i$ carrying the markup alongside it, and the dispersion we measure below contains a markup component, which Section 4.2 separates inside the model.

Misallocation. Marginal and average revenue products are proportional under (26), so equating average products is the efficiency condition: where two firms differ, moving a unit of customer capital to the higher one raises output. Holding the total stock fixed costs nothing, because customer capital enters (4) only through normalized weights and only its distribution across firms matters. Appendix A.5 keeps the structural and total

elasticities apart and reports the share response that separates them.

The efficient allocation follows from (26): equating revenue products makes the efficient stock a power function of revenue productivity, $\log b_i^e \propto \frac{1-\alpha_L}{1-\alpha_L-\alpha_B} \log \tilde{a}_i$. The gain from moving to it is

$$\log \frac{Y^e}{Y} = \frac{\alpha_B}{2(1-\alpha_L-\alpha_B)} \text{Var}_b(\log R_i - \log b_i). \quad (28)$$

Here Y is aggregate output at the observed allocation and Y^e the same object with customer capital redistributed to the optimum.

The gain from equating the revenue product of customer capital is 52% (Table 2). It requires only the estimated total elasticity, while the calibration of Section 4.2 and the welfare accounting need the exponents behind it. Constant returns on the customer-capital–labor bundle close the system: given σ , the estimated $\bar{\alpha}_B$ delivers the labor exponent ω and the demand translation ν of Table 3. The closure, rather than any one parameter, is what the evidence bears on: Appendix A.5 runs an alternative route through the flexible-input cost share, which points to a labor exponent near one, well above the closure’s 0.504.

TABLE 2: STATIC MISALLOCATION OF CUSTOMER CAPITAL

	$\sigma = 4$
Total revenue elasticity of the stock, $\bar{\alpha}_B$	0.598 (0.025)
Share response, $d \log(1 - s_i) / d \log b_i$	−0.021 (0.005)
Wedge variance, $\text{Var}_b(\log R_i - \log b_i)$	0.560
sd $\log b_i$, sd $\log \ell_i$	1.392, 1.399
corr($\log b_i, \log \ell_i$)	0.818
Gain from equating the revenue product of the stock	52%

Notes: Compustat 2014–2022, 14,288 firm-years, decay 0.15. $\bar{\alpha}_B$ is the firm-fixed-effects slope of $\log R$ on $\log b$ within industry-year; moments are under stock-share weights. The gain is (28), at fixed total stock, productivity, and labor, evaluated at the structural exponents the closure implies, $\alpha_B = \nu/\sigma = 0.372$ and $\alpha_L = 0.378$; in log points the gain is 41.9.

Customer capital and labor disperse almost identically across firms (Table 2), so the loss comes from how loosely a firm’s stock tracks the productivity behind it; how much of that gap the dynamics sustain is the question of the next section.

4.2 Dynamic Misallocation

We estimate the dynamic block by matching its five parameters, the productivity and demand dynamics (ρ_a, ν_a, ν_b) and the marketing cost (κ, θ) , to five moments of the panel, and then use the calibrated block to ask what sustains the static loss of Section 4.1.

Parameter identification. The block consists of two laws of motion. Productivity mean-reverts,

$$\ln a_{it} = \rho_a \ln a_{i,t-1} + \varepsilon_{it}^a, \quad (29)$$

while customer capital accumulates through the marketing the firm buys,

$$\ln b_{it} = \ln b_{i,t-1} + \eta_{i,t-1} + \varepsilon_{it}^b, \quad (30)$$

a random walk with an endogenous drift: η is the equilibrium marketing policy, and ε^b collects the drivers of customer capital the firm does not purchase.

The stock the data measure is built from marketing alone, while customers also arrive through channels the firm does not purchase, such as word of mouth. The perpetual inventory reproduces the investment term η exactly, so each year the constructed stock misses exactly the unpurchased shock, and the revenue residual of Section 4.1 absorbs the missed shocks as they accumulate, $\log \tilde{a}_{it} = \log a_{it} + \sum_{s \leq t} \varepsilon_{is}^b$. In growth only the current shock remains,

$$\Delta \log \tilde{a}_{it} = \Delta \log a_{it} + \varepsilon_{it}^b, \quad (31)$$

so the variance of the unpurchased shock is identified from the dynamics of measured productivity.

The three moments for the productivity process are the autocovariances of measured productivity growth, $\gamma_k = \text{Cov}(\Delta \log \tilde{a}_{it}, \Delta \log \tilde{a}_{i,t-k})$; differencing removes all permanent firm heterogeneity. The two terms of (31) separate because the ε^b are uncorrelated draws: they add to the variance γ_0 and to none of the lagged autocovariances, while the mean-reverting productivity component moves both. The lags therefore identify the productivity process on its own: the decay across lags is ρ_a , and the level of the lags gives ν_a . The variance then identifies ν_b as a leftover, the growth variance the productivity process cannot account for. Matching the autocovariance function, shown against the data out to five lags in panel (b) of Figure 3, gives the estimates of Table 3. The lags

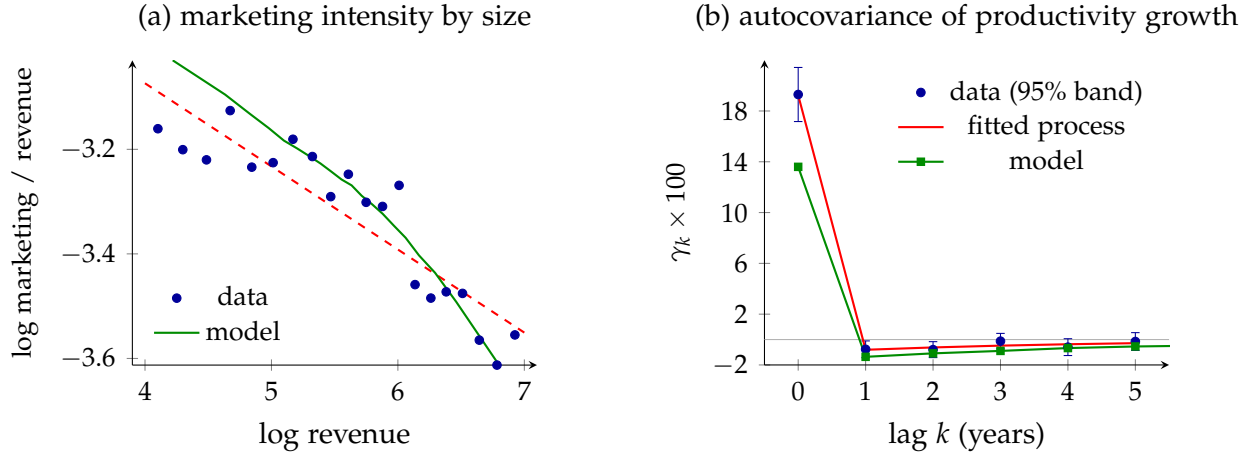
pin ρ_a loosely and v_b moves against it, so we report v_b at the matched ρ_a . About two thirds of the variance of measured productivity growth is unpurchased accumulation, the input-side counterpart of the scanner finding that firms command demand beyond what their spending built (Appendix C). The dispersion of marketing among firms of the same revenue gives an independent reading of the same shock, since conditional on revenue the composite is fixed and marketing varies only with b .

The fourth moment is marketing intensity, the marketing bill over revenue, $s_m = 0.042$ in the panel, and it identifies the cost level κ . The intensity is not an accounting identity in the parameters: the bill is the replacement flow plus net investment plus the curvature term, and the net investment rate is a choice that falls as κ rises, so the level of intensity measures how expensive accumulation is.

The fifth moment is the gradient of intensity in firm size, $d \log(m_i/R_i)/d \log R_i = -0.216$, the downward-sloping profile in panel (a) of Figure 3, which identifies the tilt θ , the exponent that scales the adjustment cost with the firm's demand weight. The tilt decides who accumulates cheaply at the margin, the cost-side counterpart of the size incentive of Section 2: a cost proportional to the stock hands the leadership gap the stock's own unit root, while a tilt against large firms makes the gap mean-revert, so the gradient moment disciplines both the cross-section of investment and the persistence of leadership at once. The calibrated $\theta > 0$ is a cost disadvantage of size: large firms pay more at the margin to grow the stock, which is what makes marketing per revenue dollar fall with size and the leadership gap mean-revert.

The cost pair that matches the two marketing moments at $I = 5$ is $\kappa = 4.4$ and $\theta = 0.80$, both moments within solver noise (Table 3). We compute every model moment with the data's own procedure: we simulate the ergodic panel, build the perpetual-inventory stock inside it from the model's measured marketing, construct $\log \tilde{a} = \log R - \bar{\alpha}_B \log \hat{b}$ with an in-panel firm-effect estimate of the total elasticity, and run the estimators of this section on the result, so the markup loads into measured productivity and every stock moment is computed on the constructed series, never on the true state. We also report one untargeted moment: the persistence of the constructed stock comes out below the data, a gap the current specification does not close, and we report it rather than absorb it.

FIGURE 3: CALIBRATION TARGETS: INTENSITY BY SIZE AND THE DYNAMICS OF MEASURED PRODUCTIVITY



Notes: Panel (a): binscatter (twenty equal-sized bins) of log marketing intensity $\log(m_i/R_i)$ on log revenue, both demeaned within NAICS3 $\times$ year cells. Panel (b): autocovariances γ_k of measured productivity growth $\Delta \log \tilde{a}_i$, firm-level means with 95% firm-clustered bands, against the fitted autocovariances of (31) (red) and the simulated game's counterparts (green). Source: model computations and Compustat/Capital IQ panel of Section 2.

Sources of misallocation. The measurement of Section 4.1 holds exactly inside the model. In the Cournot equilibrium, revenue satisfies $\log R_i = \bar{\alpha}_B \log b_i + \log \tilde{a}_i$ with b_i the physical customer-capital stock, and the revenue-productivity index conflates true productivity with the oligopoly markup, $\log \tilde{a}_i = (\tilde{\sigma}/\sigma) \log a_i + (\tilde{\sigma} - 1) \log(1 - s_i)$, as in Section 4.1. The efficient benchmark reallocates the aggregate stock each period along the log-linear line $\log b_i^* \propto \log a_i$. The static loss splits into the markup wedge and a demand misallocation term proportional to the variance of the distance from that line, which is the variance of the log revenue product we measure in the data.

The calibrated block splits the static loss of Section 4.1 into three parts, each defined by a row of Table 4. The adjustment-cost part is the loss that survives when investment is aligned but slow: with $\kappa > 0$ the stock tracks productivity with a lag, so dispersion in the revenue product remains even when every firm invests toward the efficient line, and the distance between free marketing and the first-best planner measures it. The legacy part is the loss the equilibrium adds on top: the tilt and the strategic value of accumulated demand reward size regardless of productivity, so firms defend stocks the efficient allocation would move, and the first-best planner, who corrects the incentive but pays the same adjustment cost firms do, recovers exactly this part. Adjustment costs account for 44%, the legacy effect for 11%, and a residual from outside the model

TABLE 3: CALIBRATION: MOMENT MATCHING

<i>A. Parameters</i>			
		Identification	Value
Substitution elasticity	σ	Hottman et al. (2016)	4
Labor elasticity	ω	constant returns, given $\sigma, \bar{\alpha}_B$	0.504
Customer-capital elasticity	ν	constant returns, given $\sigma, \bar{\alpha}_B$	1.488
PIM depreciation	δ	useful-life estimates	0.15
Stock share	s_b	measured $\sum_i b_i / R$	0.21
Discount factor	β	annual discounting	0.96
Number of firms	I	effective competitors	5
Productivity persistence	ρ_a	decay of γ_k across lags	0.771
Productivity shock	ν_a	level of $\gamma_k, k \geq 1$	0.622
Demand shock	ν_b	growth variance γ_0 , as leftover	0.868
Marketing cost level	κ	marketing intensity s_m	4.4
Marketing cost tilt	θ	intensity-size gradient	0.80

B. Moments

	Model	Data
Growth variance (targeted)	19.25	19.30
Autocovariance $t, t - 1$ (targeted)	-0.81	-0.77
Autocovariance $t, t - 2$ (targeted)	-0.62	-0.79
Marketing intensity s_m (targeted)	0.043	0.042
Intensity-size gradient (targeted)	-0.212	-0.216
Constructed-stock persistence (untargeted)	0.911	0.964

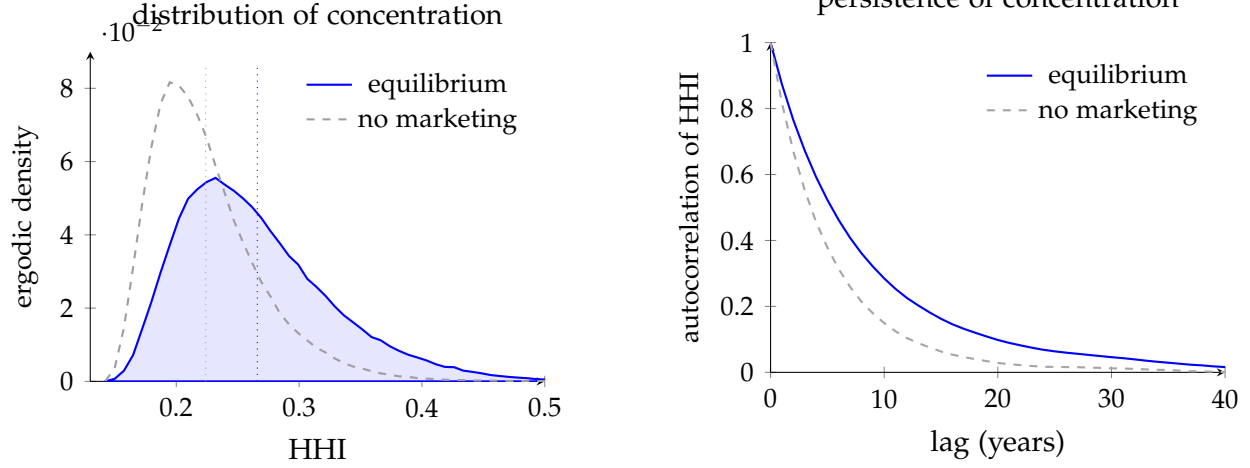
Notes: Constant returns close the static block: given σ , the estimated $\bar{\alpha}_B$ delivers ω and ν , with $\bar{\sigma} = 1.608$ of (12). The two shock rows are in state units, $\sigma/\bar{\sigma}$ times their $\bar{\alpha}$ -unit estimates of 0.250 and 0.349. Source: model computations.

generates 45%, as shares of the 41.9 log points that the 52% gain of Section 4.1 represents. The legacy share is 4.7 log points of aggregate productivity, and because the planner rows of Table 4 use linear policy rules it is a lower bound.

4.3 Concentration and Productivity

Marketing raises both the level and the persistence of concentration. Against a no-marketing benchmark simulated on the same shock draws, the equilibrium raises expected concentration from an HHI of 0.224 to 0.266, moving the effective number of firms from 4.6 to 4.0, which the calibration never targeted. The larger change is dynamic: the closed marketing loop raises the annual persistence of the leadership gap from 0.92 to 0.95, a half-life of 14.1 rather than 8.2 years. This wedge is the legacy effect of Proposition 2 at the calibrated parameters, the same force the decomposition of Section 4.2 prices at 4.7 log points. Figure 4 displays both margins, the rightward shift of the ergodic concentration distribution and the slower decay of its autocorrelation.

FIGURE 4: CONCENTRATION IN THE CALIBRATED GAME AND WITHOUT MARKETING



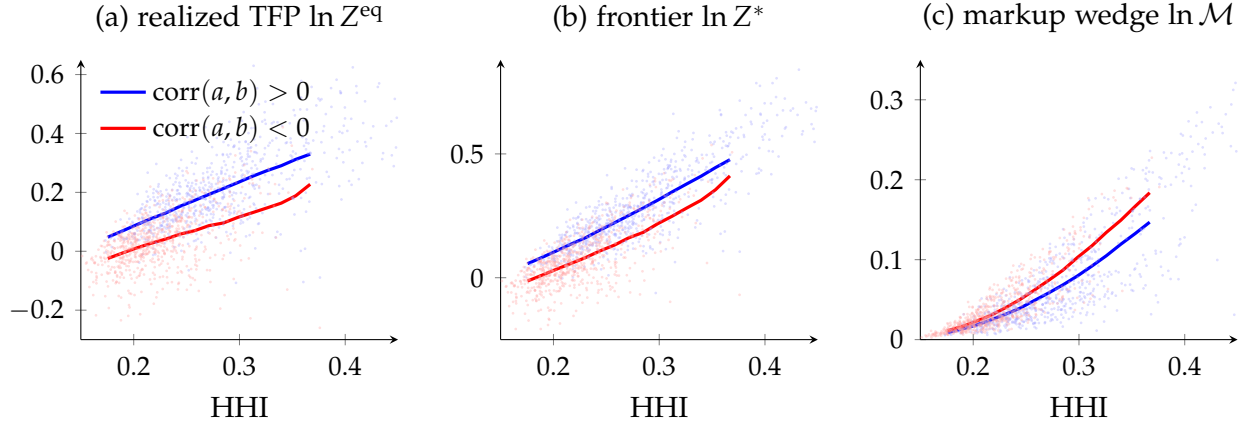
Notes: Calibrated ergodic panel and the no-marketing benchmark on common shock draws ($T = 200,000$). Left: ergodic density of the market HHI; dotted lines mark the means (0.224 without marketing, 0.266 in equilibrium). Right: autocorrelation of the HHI by lag. Source: model computations.

The cross-sectional relationship between concentration and productivity has no fixed sign, as the sign test of Corollary 1 already showed at two firms. At $I = 5$ the same pattern holds state by state (Figure 5): in states where the productive firms hold the demand, realized TFP rises with concentration, and in misaligned states it falls, while the markup component rises with concentration in both. The ergodic mean of the alignment is $+0.21$, so the economy's average slope is positive, again untargeted. Grouping states by their productivity dispersion instead isolates the same margin from the productivity side (Figure 6): the more productivity dispersion a state has, the steeper its concentration–productivity slope. This positive analysis suggests the costs of concentration are closely linked to the drivers; when markets have high and persistent productivity dispersion, concentration tends to be a much more benign outcome than when dispersion is low.

5 Counterfactuals and Policy

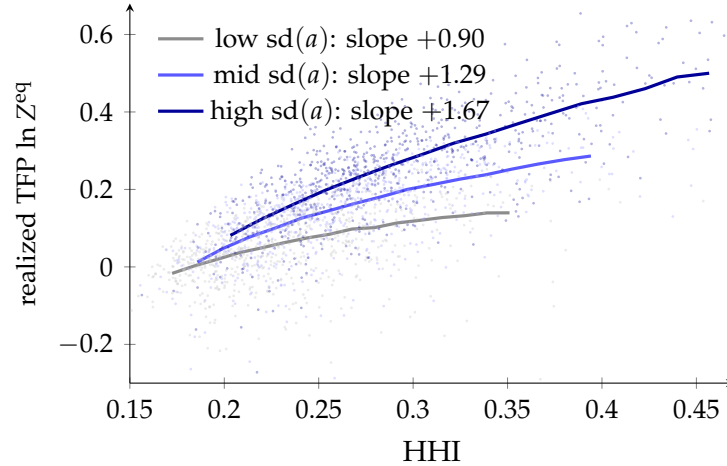
We use the calibrated model to measure what competition in demand costs the household and what a policymaker could recover. We characterize the planner's solution, compare the equilibrium to four counterfactuals on common shock draws (shutting marketing down, free marketing, a constrained planner under Cournot pricing, and the first

FIGURE 5: CONCENTRATION AND PRODUCTIVITY IN THE CROSS-SECTION



Notes: Calibrated model at $I = 5$, ergodic states ($T = 200,000$) split by the sign of the within-state cross-firm correlation of productivity and customer capital; thick lines are conditional means by HHI. Realized TFP (a) decomposes into the frontier (b) net of the markup wedge (c), $\ln Z^{\text{eq}} = \ln Z^* - \ln \mathcal{M}$ (Lemma 1). Source: model computations.

FIGURE 6: CONCENTRATION AND PRODUCTIVITY, STATES GROUPED BY PRODUCTIVE DISPERSION



Notes: Ergodic states grouped into terciles of the within-state cross-firm standard deviation of productivity; thick lines are conditional means by HHI, legend slopes from a linear fit of $\ln Z^{\text{eq}}$ on HHI within each tercile. Source: model computations.

best), and close with a size-based subsidy that backfires.

5.1 Planner's Problem

The planner chooses every firm's marketing at once, under one of two static regimes. The *constrained planner* inherits Cournot pricing and controls marketing only, so its gap

to the equilibrium isolates the conduct of the marketing game; the *first-best planner* also allocates production labor, so its additional gap measures market power itself. Formally, for regime $r \in \{C, F\}$ the planner solves

$$V^P(\mathbf{x}) = \max_{\mathbf{m} \geq \mathbf{0}} \left\{ W^r(\mathbf{x}) - \sum_{i=1}^I K_i(\eta_i, \mathbf{b}) + \beta \mathbb{E}[V^P(\mathbf{x}') \mid \mathbf{x}, \mathbf{m}] \right\}, \quad (32)$$

subject to the laws of motion (6) and (7) for every firm, where $K_i(\eta_i, \mathbf{b}) = m_i + \frac{\kappa}{2} \left(\frac{b_i}{\sum_j b_j} \right)^\theta \eta_i^2$ is the marketing bill of (16) at the rate $\eta_i = m_i / b_i$, with the regime flows

$$W^C(\mathbf{x}) = \ln C^{\text{eq}}(\mathbf{x}) - L^{\text{prod}}(\mathbf{x}), \quad (33)$$

$$W^F(\mathbf{x}) = \omega \ln \omega - \omega + \ln Z^*(\mathbf{x}). \quad (34)$$

The objective is household welfare, the flow (23) with the marketing bill written out, evaluated at the regime's statics. The constrained flow W^C is the household flow at the Cournot allocation; the first-best flow is its value with production labor allocated efficiently, $\max_{\ell, L} \{\ln C - L\} = \omega \ln \omega - \omega + \ln Z^*$ (Appendix A.1). The two regimes bracket what policy can reach: conduct remedies alone, and conduct together with the output-side correction.

Level invariance makes the social value of customer capital zero-sum. A uniform scaling of $\mathbf{b}$ leaves the allocative part of welfare unchanged, so the social flow value sums to zero across firms at every state, and the same holds for the discounted shadow values (Appendix A.4). The planner's marketing is therefore purely redistributive: negative marketing on one firm offsets positive marketing on another one-for-one, and at any symmetric state the planner does not market at all. In the equilibrium, by contrast, each firm's private marginal value of customer capital at the symmetric state is 0.043 while the social value is exactly zero, so marketing at the modal state redistributes customers without creating surplus.

The planner's first-order condition clarifies what a corrective policy must change. The planner is a single agent, so the envelope theorem removes the strategic-feedback

block from the shadow-value recursion:

$$\eta_i^P(\mathbf{x}) = \max \left\{ \frac{b_i}{\kappa} \left(\frac{b_i}{\sum_j b_j} \right)^{-\theta} \left(\beta \mathbb{E} \left[\frac{\partial V^P}{\partial b'_i}(\mathbf{x}') \varepsilon_i^b \right] - 1 \right), 0 \right\}, \quad (35)$$

$$\frac{\partial V^P}{\partial b_i}(\mathbf{x}) = g_i(\mathbf{x}) - \frac{\partial}{\partial b_i} \sum_k K_k + (1 - \delta) \beta \mathbb{E} \left[\frac{\partial V^P}{\partial b'_i}(\mathbf{x}') \varepsilon_i^b \right], \quad (36)$$

the same pair as the equilibrium conditions (17) and (18), with the private flow $\partial \pi_i / \partial b_i$ replaced by the social flow g_i , every firm's bill internalized, and no strategic-feedback term. In the first-best regime the social flow has a closed form,

$$g_i^F = \frac{\partial \ln Z^*}{\partial b_i} = \frac{\nu}{(\sigma - 1) b_i} \left[\frac{(a_i b_i^\nu)^{\tilde{\sigma}/\sigma}}{\sum_j (a_j b_j^\nu)^{\tilde{\sigma}/\sigma}} - \frac{b_i^\nu}{\sum_j b_j^\nu} \right], \quad (37)$$

positive exactly when firm i 's frontier weight exceeds its customer-capital weight, so marketing has social value only when it moves demand toward firms whose composite productivity is high relative to the stock they already hold. Appendix A.4 derives the constrained-regime flow in closed form and verifies both flows against finite differences.

5.2 Implementing the Planner Solution

The planner's problem solves with the equilibrium's computational machinery: under the Stein representation it is the critic-actor pair

$$\eta_i^P(\mathbf{x}) = \max \left\{ \frac{b_i}{\kappa} \left(\frac{b_i}{\sum_j b_j} \right)^{-\theta} \left(\frac{\beta}{\nu_b^2} \frac{\mathbb{E}[\ln \varepsilon_i^b V^P(\mathbf{x}') | \mathbf{x}]}{(1 - \delta) b_i + m_i} - 1 \right), 0 \right\}, \quad (38)$$

$$V^P(\mathbf{x}) = W^r(\mathbf{x}) - \sum_{i=1}^I K_i(\eta_i^P(\mathbf{x}), \mathbf{b}) + \beta \mathbb{E}[V^P(\mathbf{x}') | \mathbf{x}], \quad (39)$$

the same operator as the equilibrium system (19), solved by the same family and algorithm on the same laws of motion (6) and (7). The distance between the two problems has an exact expression in terms of the total private value.

Definition 2 (Total private value) $V^\Sigma(\mathbf{x}) \equiv \sum_{i=1}^I V_i(\mathbf{x})$, the sum of the firms' equilibrium values.

Summed across i , the firms' critics (19) yield a total that obeys the planner's recursion

(39) with aggregate profit as its flow,

$$V^\Sigma(\mathbf{x}) = \underbrace{\frac{1}{\tilde{\sigma}} + \frac{\tilde{\sigma}-1}{\tilde{\sigma}} \text{HHI}(\mathbf{z})}_{\text{aggregate profit } \Pi} - \sum_{i=1}^I K_i(\eta_i(\mathbf{x}), \mathbf{b}) + \beta \mathbb{E}[V^\Sigma(\mathbf{x}') | \mathbf{x}], \quad (40)$$

with the full marketing bill in both. With labor in closed form ($L^{\text{prod}} = 1/\bar{\mu}$, Lemma 1), aggregate profit is linear in concentration with slope $\frac{\tilde{\sigma}-1}{\tilde{\sigma}}$. Two wedges separate the equilibrium from the planner. In the actor, each firm receives only its own component of V^Σ , so it ignores the value its marketing destroys at rivals. This is the *business-stealing wedge*, and by the zero-sum result above it accounts for all marketing at symmetric states. In the flow, the constrained-social objective is $W^C = \ln C^{\text{eq}} - \frac{\tilde{\sigma}-1}{\tilde{\sigma}}(1 - \text{HHI})$; its concentration term cancels against the one in Π , leaving real consumption alone, $W^C - \Pi = \ln C^{\text{eq}} - 1$. This *valuation wedge* is the consumer surplus firms cannot collect. The strategic feedback in (18) is the shadow-value counterpart of the first wedge, and the flow gap $g_i - \phi_{ii}$ nets both; the whole correction therefore collapses into a single marketing-side schedule.

Lemma 2 (A marketing subsidy implements the planner) *For either regime $r \in \{C, F\}$ there is a state-dependent subsidy $\tau_i(\mathbf{x})$ per unit of marketing, financed lump sum, such that the planner's policies form a marketing equilibrium of the subsidized game, in which firm i 's flow is $\pi_i + \tau_i(\mathbf{x}) \eta_i - K_i(\eta_i, \mathbf{b})$. The schedule equals the sum of the two wedges on the implemented path,*

$$\tau_i(\mathbf{x}) = \underbrace{\frac{\beta}{v_b^2} \mathbb{E}^P[\varepsilon_i^b (V^P - \tilde{V}^\Sigma)(\mathbf{x}') | \mathbf{x}]}_{\text{valuation}} + \underbrace{\frac{\beta}{v_b^2} \sum_{k \neq i} \mathbb{E}^P[\varepsilon_i^b \tilde{V}_k(\mathbf{x}') | \mathbf{x}]}_{\text{business stealing}}, \quad (41)$$

where $\tilde{V}_k$ is firm k 's value in the subsidized game and $\tilde{V}^\Sigma$ their total; no re-equilibration term appears, because the continuation is the planner's own.

The derivation is in Appendix A.4. The instrument is marketing-side alone, leaving pricing, production, and the laws of motion untouched. Each component of (41) is a closed-form evaluation of fitted critics, so the two wedges can be drawn state by state; Section 5.4 does so after the counterfactual numbers.

5.3 Counterfactual Welfare

We perform four counterfactuals to study the process. Shutting marketing down ($\eta \equiv 0$) keeps Cournot pricing and lets both stocks evolve on their own. Free marketing is the frictionless benchmark: customer capital is reassigned each period to the efficient line $\ln b_i^* = \frac{\tilde{\sigma}}{\sigma(1-\tilde{\alpha}_B)} \ln a_i$ of (A15) at no resource cost, with labor allocated efficiently. The two planner rows are the regimes of the previous subsection, feasible marketing policies under Cournot pricing and under efficient labor allocation.

TABLE 4: COUNTERFACTUALS: WELFARE RELATIVE TO THE EQUILIBRIUM

	ΔW	$\Delta \ln Z^*$	$-\Delta(\ln \mathcal{M} + \Lambda)$	$-\Delta L^{\text{mkt}}$	ΔHHI	$\Delta \text{corr}(a, b)$
equilibrium	0	0	0	0	0	0
$\eta \equiv 0$	+5.2	-4.9	+5.0	+5.1	-.041	-.21
free marketing	+50.2	+23.1	+22.0	+5.1	+.072	+.79
constrained planner	+8.8	+0.9	+3.3	+4.6	-.027	+.06
first-best planner	+30.4	+4.7	+22.0	+3.7	+.001	+.12

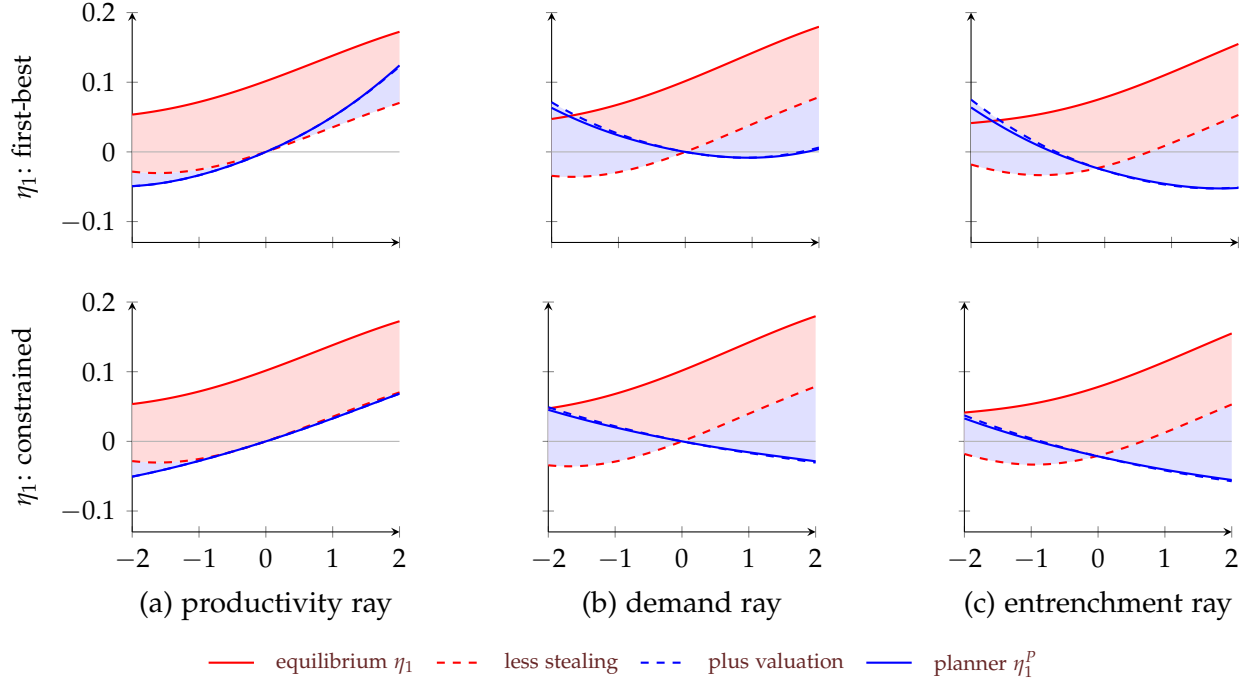
Notes: Entries are welfare changes in log points of flow utility, consumption-equivalent percent under log utility; the three split columns (the frontier, the net cost of market power, and the marketing bill) sum to ΔW state by state, by the identity (23). Ergodic means over 2×10^5 states at the calibrated parameters, common shock draws across scenarios.

Shutting marketing down raises welfare by 5.2 percent (Table 4, $\eta \equiv 0$): the recovered bill (5.1) and the market-power relief (5.0) outweigh the lost alignment (4.9 points of frontier). The constrained planner does better by allowing some marketing. Its total gain is at least 8.8 points, at least 3.6 above the marketing ban. Redirecting rather than banning marketing turns the frontier loss into a 0.9-point gain and raises the alignment $\text{corr}(a, b)$ by 0.06. Its policy is close to linear in the state: marketing rises with relative productivity and falls with inherited customer capital.

5.4 Corrective Wedges

The correction is largest in states where demand is inherited. Figure 7 plots the equilibrium marketing rate against the planner's along three state rays, the first-best regime in the top row and the constrained regime beneath, with the gap split into the two wedges of Lemma 2. In every panel the red band removes business stealing and the blue band then adds the valuation wedge, landing on the planner's rate. At the symmetric state the fitted solutions reproduce the zero-sum result exactly. Both planners stop marketing, and all marketing is business stealing. Along the productivity ray, the planner keeps

FIGURE 7: THE CORRECTION IN INVESTMENT RATES: EQUILIBRIUM VERSUS PLANNER



Notes: Firm 1's fitted marketing rate in the equilibrium (red) and under the planner (blue) along state rays from the symmetric state, top row first-best, bottom row constrained; horizontal axes: the ray deviation, (a) a_1 pushed, (b) b_1 pushed, (c) b_1 pushed with $a_1 < 0$. Red band: the business-stealing component of the gap; blue band: the valuation component; dashed: the equilibrium rate with each component applied in turn. Fitted solutions at the calibrated parameters. Source: model computations.

marketing the productive leader at a rate below the equilibrium's by roughly the stealing wedge, which is nearly flat across states. Along the demand and entrenchment rays, the equilibrium raises its marketing rate with the accumulated advantage while both planners set negative marketing for the entrenched incumbent.

The correction is therefore (a, b) -dependent, and an instrument must tell the two dimensions apart to implement it. The planner conditions on the split directly; a policymaker who observes only size cannot, and any payment that conditions on size rewards the composite, accumulated stock included. The next subsection shows what that constraint costs.

5.5 Size-Based Policy

Now consider a policymaker who observes size but not its source, and who pays a size-based output subsidy offsetting a fraction $\lambda \in [0, 1]$ of each firm's log markup (construction and closed forms in Appendix A.7). At $\lambda = 0$ the subsidy is off; at $\lambda = 1$

pricing is at marginal cost and the statics are the first-best allocation given $\mathbf{z}$, the same statics the first-best planner implements. The static allocation improves by construction at every λ , so the question is how the marketing game responds to the reshaped incentives. Every λ -economy is solved at the same discount $\beta = 0.96$ as the equilibrium, on common shock draws.

At $\lambda = 1$ welfare falls 16.0 points relative to the equilibrium.¹¹ The static gain is 22.0 points, but the subsidy rewards size, and firms acquire size through marketing: the private value of customer capital rises by half, the marketing bill triples, the frontier falls, and effective firms drop from four to one and a half.

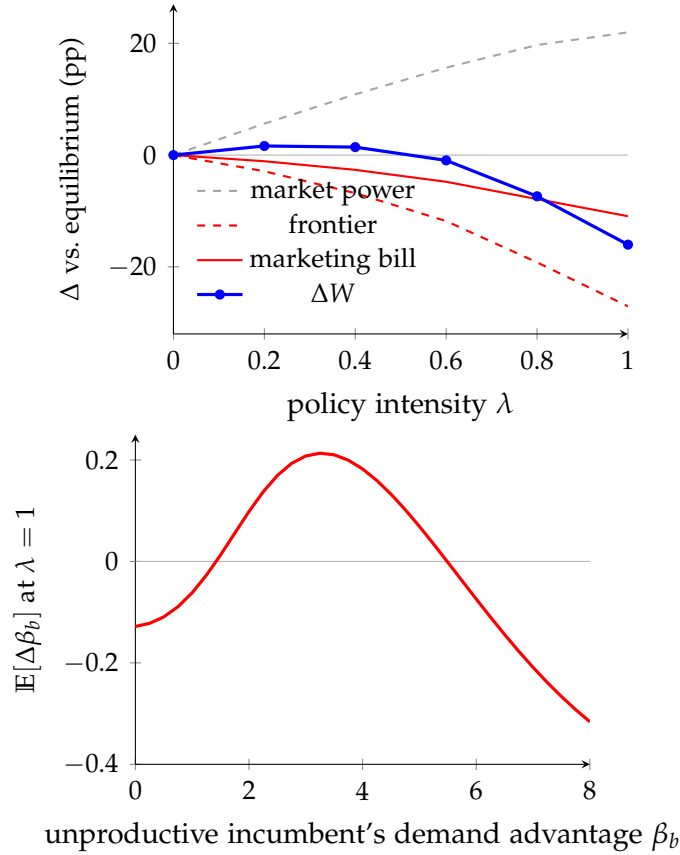
Along the λ -path (Figure 8, left panel) the welfare change peaks near $\lambda = 0.2$ at +1.6 points, crosses zero near $\lambda = 0.55$, and falls steeply thereafter as intensified marketing spending consumes it. The optimal λ corrects a fifth of the markup. The right panel sets $\lambda = 1$: for an unproductive incumbent, the expected drift of its demand advantage turns positive over a wide range, so the subsidy sustains leads that the laissez-faire equilibrium never generates.

The planner's policy locates the social problem in the *size incentive* rather than in marketing itself; the planner still markets. Private returns rise with the accumulated stock while social returns fall with it, so instruments that reward size, output subsidies included, push marketing in the wrong direction, the mechanism behind the backfire above. And because the loss the game sustains is the entrenchment component of the measured misallocation, policies that only compress markups leave the dynamic distortion in place; the corrective margin is the matching between demand and productivity rather than the volume of marketing.

6 Conclusion

This paper studies market concentration through the lens of accumulated customer demand. In an oligopolistic setting, marketing can entrench incumbents: past productivity advantages leave accumulated demand that becomes self-sustaining, because firms defend the stock they hold even after the productivity behind it fades. We term this force the *legacy effect*. It disconnects firm size from productivity, and the dynamics of accu-

¹¹The marketing-bill series is expressed at the 4.2 percent data share, with the residual marketing cost folded into the frontier series; levels are not directly comparable to Table 4.

FIGURE 8: THE λ -SUBSIDY: WELFARE PATH AND ENTRENCHMENT DRIFT


Notes: Left: welfare change and its three channels along the subsidy path, ergodic means of each λ -economy against the equilibrium, common shock draws. Right: expected one-period change of an unproductive incumbent's demand advantage β_b at $\lambda = 1$. Source: model computations.

mulation sustain a significant share of the misallocation we measure between customer capital and productivity, so concentration reflects past marketing as much as current productivity. Aggregate productivity is 5% lower due to the legacy effect.

The quantitative results carry lessons for competition policy. Standard size-based policies to alleviate markups may reward accumulating demand, which can entrench incumbents; correcting this classic static markup distortion can deepen the dynamic one. Policies aimed at concentration need to register the sources of concentration. Demand-driven concentration can be costly and persistent, whereas productivity-driven concentration tends to have initially positive consequences.

This paper does not speak to the endogenous feedback loop between demand and productivity, and the many roles of customer acquisition at different stages of the firm life cycle. These are exciting areas for future research in the world of market concentra-

tion, productivity, and development. We expect the demand side of firms will remain an important element for studies on a wide range of topics.

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Appendix

The Appendix contains two sections: the Model Appendix (A) collects the model's proofs and derivations, and Appendix B covers the estimation, moment conditions, and robustness.

A Model Appendix

Notation. Δ is the difference operator throughout; the composite curvature of (12) is written $\sigma/\tilde{\sigma}$.

This appendix collects the proofs and derivations for the model of Section 3: the static-equilibrium objects (Appendix A.1), the Stein representation and the critic family behind the computational method (Appendix A.2), the two-firm results (Appendix A.3), the welfare accounting, the planner's solution, and the Hsieh–Klenow bridge (Appendix A.4), the rational-inattention microfoundation (Appendix A.6), the policy extensions (Appendix A.7), and aggregation with the household problem (Appendix A.8).

A.1 Static Equilibrium: Shares, Marginal Profits, and Output

Share Jacobian. Totally differentiating the share fixed point (13) with respect to a perturbation db_i , holding the other states constant, gives for the focal firm

$$\frac{ds_i}{s_i} = \frac{\tilde{\sigma}}{\sigma} db_i - (\tilde{\sigma} - 1) \frac{ds_i}{1 - s_i} - \sum_{\ell=1}^I s_{\ell} \left[\frac{\tilde{\sigma}}{\sigma} db_{\ell} - (\tilde{\sigma} - 1) \frac{ds_{\ell}}{1 - s_{\ell}} \right],$$

and the same expression without the db_i term for every rival. Guessing a solution proportional to the share responsiveness $\chi_i \equiv \frac{\tilde{\sigma}}{\sigma} \left(\frac{1}{s_i} + \frac{\tilde{\sigma}-1}{1-s_i} \right)^{-1}$, with normalized responsiveness $\hat{\chi}_i \equiv \chi_i / \sum_{\ell} \chi_{\ell}$, and verifying,

$$\begin{aligned} \frac{\partial s_i}{\partial b_i} &= \chi_i (1 - \hat{\chi}_i), \\ \frac{\partial s_j}{\partial b_i} &= -\chi_j \hat{\chi}_i \quad (j \neq i), \end{aligned} \tag{A1}$$

which satisfies the adding-up check $\sum_j \partial s_j / \partial b_i = \chi_i - \chi_i = 0$. The responsiveness χ_i is hump-shaped in the share: firms at intermediate shares respond most to customer capital, dominant and marginal firms least. Because profits (14) depend on the own share

alone, with marginal profit $\pi'(s) = \frac{1}{\tilde{\sigma}} + 2\frac{\tilde{\sigma}-1}{\tilde{\sigma}}s$, the marginal-profit matrix of customer capital is

$$\phi_{ik}(\mathbf{z}) \equiv \frac{\partial \pi_i}{\partial b_k} = \pi'(s_i) \chi_i (\mathbb{I}_{i=k} - \hat{\chi}_k), \quad (\text{A2})$$

the flow term of the envelope condition (18).

Labor pools. With unit expenditure, firm i 's revenue is s_i and its production labor is revenue over the average-cost markup, $\ell_i = s_i / \tilde{\mu}_i = \frac{\tilde{\sigma}-1}{\tilde{\sigma}} s_i (1 - s_i)$; summing across firms delivers $L^{\text{prod}} = 1 / \bar{\mu}$ of Lemma 1.

Equilibrium output. The starting object is (15). The pricing stage allocates $\ell_i = s_i / \tilde{\mu}_i = \frac{\tilde{\sigma}-1}{\tilde{\sigma}} s_i (1 - s_i)$, and substituting into the aggregator with $e^{z_i/\sigma} = (a_i b_i^\nu)^{1/\sigma}$,

$$C^{\text{eq}}(\mathbf{z}) = \left(\frac{\tilde{\sigma} - 1}{\tilde{\sigma}} \right)^\omega \frac{\left[\sum_j e^{z_j/\sigma} (s_j (1 - s_j))^\frac{\omega(\sigma-1)}{\sigma} \right]^\frac{\sigma}{\sigma-1}}{(\sum_k b_k^\nu)^{\frac{1}{\sigma-1}}}. \quad (\text{A3})$$

The frictionless allocation $\ell_i^* \propto e^{\hat{z}_i}$ (maximizing C at fixed total labor; the first-order condition of (15) in ℓ_i) delivers $C^* = \left[(\sum_j e^{\hat{z}_j})^{\sigma/\tilde{\sigma}} / \sum_k b_k^\nu \right]^{1/(\sigma-1)} L^\omega = Z^* L^\omega$, the frontier index Z^* of Lemma 1 times the labor level at its exponent.

A.2 Computation: Stein Representation and the Critic Family

Level representation. Stein's lemma applied to the first-order and envelope conditions (17) and (18) gives the critic-actor system (19) (the *critic* is the value recursion, the *actor* the policy condition) on the stacked state $\mathbf{x} = (\mathbf{a}, \mathbf{b})$. Higher derivatives reduce to levels in the same way, with Hermite weights, $\mathbb{E}[\partial^2 f / \partial b_k' \partial b_i' | \mathbf{x}] = \mathbb{E}[(\varepsilon_k^b \varepsilon_i^b - \nu_b^2 \mathbb{I}_{k=i}) f(\mathbf{x}') | \mathbf{x}] / \nu_b^4$; differentiating the actor equation and applying this identity shows that the strategic slopes are shock-weighted levels to leading order in the policy slopes and the cost cross-partials,

$$\frac{\partial \eta_k(\mathbf{x})}{\partial b_i} \approx \frac{\beta}{(\partial^2 C_k / \partial \eta_k^2) \nu_b^4} \mathbb{E}[(\varepsilon_k^b \varepsilon_i^b - \nu_b^2 \mathbb{I}_{k=i}) V_k(\mathbf{x}') | \mathbf{x}].$$

Interiority (the second-order condition) also takes a level form, $\frac{\beta}{\nu_b^2} \mathbb{E}[(\varepsilon_i^b)^2 / \nu_b^2 - 1] V_i(\mathbf{x}') | \mathbf{x}] < \partial^2 C_i / \partial \eta_i^2$, which we verify at every solution. Because $\mathbf{x}'$ contains $\boldsymbol{\eta}(\mathbf{x})$, the action vector at each state solves a small fixed point per evaluation; the squared policy term keeps the pair from being a monotone contraction, so we iterate a damped critic-actor alternation

under a discount-continuation schedule.

Index-function reduction. Let T_w denote the Gaussian smoothing operator, $(T_w f)(y) \equiv \mathbb{E}[f(y + w\tilde{\zeta})]$ with $\tilde{\zeta} \sim N(0, 1)$, which satisfies the semigroup property $T_w \circ T_{w'} = T_{\sqrt{w^2 + w'^2}}$. For a direction $d = (d^a, d^b)$, mean zero within each stock, and an absolutely continuous profile f of polynomial growth, the index $d \cdot \mathbf{x}'$ is univariate Gaussian conditional on $\mathbf{x}$, $d \cdot \mathbf{x}' \sim N(d \cdot \mathbf{x}^e, v_d^2)$ with the expected next state $\mathbf{x}^e(\mathbf{x}) \equiv (\rho_d \mathbf{a}, \mathbf{b} + \boldsymbol{\eta}(\mathbf{x}))$ and shock loading $v_d^2 \equiv v_a^2 |d^a|^2 + v_b^2 |d^b|^2$, and

$$\begin{aligned} \mathbb{E}[f(d \cdot \mathbf{x}') | \mathbf{x}] &= (T_{v_d} f)(d \cdot \mathbf{x}^e), \\ \mathbb{E}[\varepsilon_k^b f(d \cdot \mathbf{x}') | \mathbf{x}] &= v_b^2 d_k^b (T_{v_d} f)'(d \cdot \mathbf{x}^e). \end{aligned} \quad (\text{A4})$$

Hence if the profile family is closed under T_w with an explicit parameter map (*conjugacy*), every expectation entering the critic and the actor is an evaluation of a member of the same family, and the multivariate problem requires only this one-dimensional property. Polynomials are conjugate through Gaussian moments (degree two is the linear-quadratic block, which serves as the warm start), and probits are conjugate through cdf smoothing,

$$\begin{aligned} \mathbb{E}\left[\Phi\left(\frac{d \cdot \mathbf{x}' - \bar{x}}{w}\right) | \mathbf{x}\right] &= \Phi\left(\frac{d \cdot \mathbf{x}^e - \bar{x}}{\sqrt{w^2 + v_d^2}}\right), \\ \mathbb{E}\left[\varepsilon_k^b \Phi\left(\frac{d \cdot \mathbf{x}' - \bar{x}}{w}\right) | \mathbf{x}\right] &= \frac{v_b^2 d_k^b}{\sqrt{w^2 + v_d^2}} \Phi'\left(\frac{d \cdot \mathbf{x}^e - \bar{x}}{\sqrt{w^2 + v_d^2}}\right), \end{aligned} \quad (\text{A5})$$

so with the family (22) the actor is an explicit function of the coefficients. The derivative of a probit is a Gaussian density, which is how the sigmoid terms of the value function generate the hump-shaped policies of the two-firm analysis.

Symmetry and orbit-tying. The primitives are exchangeable, so symmetric-equilibrium values are equivariant, $V_{\varrho(i)}(\varrho \cdot \mathbf{x}) = V_i(\mathbf{x})$ for every permutation ϱ of the firms (acting on both stocks at once); equivariance across firms is imposed by construction (firm j 's members are firm i 's with indices transposed), leaving the rival-permutation invariance $V_i(\varrho \cdot \mathbf{x}) = V_i(\mathbf{x})$ for ϱ fixing i . A single index member $f(d \cdot \mathbf{x})$ with non-constant f satisfies this only if the rival weights of d are all equal (the own-index direction) or the orbit is a reflection pair with f even (the rival-gap quadratic), because invariance forces

$f(d \cdot \mathbf{x}) = f(d' \cdot \mathbf{x})$ for every d' in the orbit, two non-parallel directions would force f to be constant, and permutations preserve the norm. Free directions therefore enter only through the group average over rival permutations,

$$G_d(\mathbf{x}) = \sum_q f(d_q \cdot \mathbf{x}), \quad (\text{A6})$$

the orbit sum with one tied coefficient. The orbit sum is invariant by construction; conjugacy is preserved because $|d_q| = |d|$, so the smoothing acts term by term at a common width and commutes with the symmetrization; and no flexibility is lost, because the group average is a norm-nonexpansive linear projection onto the invariant functions, so density of the untied family implies density of the tied family in the invariant class. The main pattern we use is the distinguished-rival direction (own weight, one rival's weight, and a common weight on the rest), whose orbit has size $I - 1$, so its cost is linear in I ; it covers the firm's own position, each bilateral rivalry, and the rest-of-market aggregate.

Fitting as regression. The coefficients are fitted to *policy-relevant* targets rather than to the Bellman residual: at each sampled state, common-random-number finite differences of the simulated discounted return in the own customer-capital direction estimate the marginal value the first-order condition prices,

$$\hat{q}_i(\mathbf{x}) = \frac{1}{2h} \mathbb{E} \left[R_i^{(+h)}(\mathbf{x}) - R_i^{(-h)}(\mathbf{x}) \right], \quad R_i^{(\pm h)} = \sum_{t=0}^{T-1} \beta^t \text{flow}_{i,t}^{(\pm h)} + \beta^T V_{\boldsymbol{\theta}}(\mathbf{x}_T^{(\pm h)}), \quad (\text{A7})$$

with the perturbation $\pm h$ applied to b_i at the first transition, all shock draws common across the pair, trimmed means across paths, and the tail closed by the current critic (the one acknowledged approximation, with weight β^T). Because the fitted $q_{\boldsymbol{\theta}}$ implied by the family (22) is linear in $\boldsymbol{\theta}$, matching $\hat{q}$ together with the mean returns (which pin the level) is one weighted least-squares solve per round, with a ridge toward the incumbent coefficients and a trust region on the symmetric-state q ; no multi-start and no nonlinear search are needed. The steps are: (1) start from the symmetric replacement seed (the linear member carrying K' at zero net investment); (2) simulate an ergodic panel under the current actor; (3) form the targets (A7) on the panel plus dilated off-manifold states; (4) update $\boldsymbol{\theta}$ by the weighted least squares and repeat from (2) until the sample and coefficients are jointly stable; (5) audit: the target residual, the distance to the exact small- I solver, the interiority check, and the stationarity certificate on every pairwise gap. The smallest M that passes the audit measures the equilibrium's nonlinearity, and

each additional member adds one sigmoidal component to the value function.

Conjugate families and the basis used. Other families are also conjugate under T_w but less useful in practice: exponentials are ill conditioned on wide ergodic supports, and rank-based symmetrizations are not conjugate, because sorting is only piecewise linear in the shock. The calibrated solves use the linear-quadratic block on the own- and rival-mean indices of each stock together with orbit-tied probit members (A6) on the pairwise composite gaps $(z_i - z_j)/2$ (the distinguished-rival pattern), adding members while the out-of-sample residual remains large.

Approximation accuracy. The audit against exact solutions runs on a test game with a common retention ($\sigma = 3.9$, $\rho_a = 0.87$, $\kappa = 1.1$, $\nu_a = 0.1$), where exact solvers exist at small firm counts: a one-dimensional gap solver at $I = 2$ and a two-dimensional collocation solver at $I = 3$ (its own permutation-symmetry error 2.7×10^{-4}). Table A1 reports the fitted family against these benchmarks and its out-of-sample Bellman residuals through $I = 5$; Figure A1 overlays the fitted and exact policies at $I = 2$ and traces the out-of-sample residual as probit members are added. The fitted policies are within a few parts in ten thousand of the exact ones, and the residual flattens after two members, so the measured nonlinearity of this game is $M = 2$. At $\beta = 0.95$ the audits pass at $I = 2$ and 3 while the closed loop nears a unit root at $I \geq 4$, where the stationarity certificate is the binding audit; the calibrated solves on the perpetual-inventory technology at $\beta = 0.96$ report their own out-of-sample residuals with each results table.

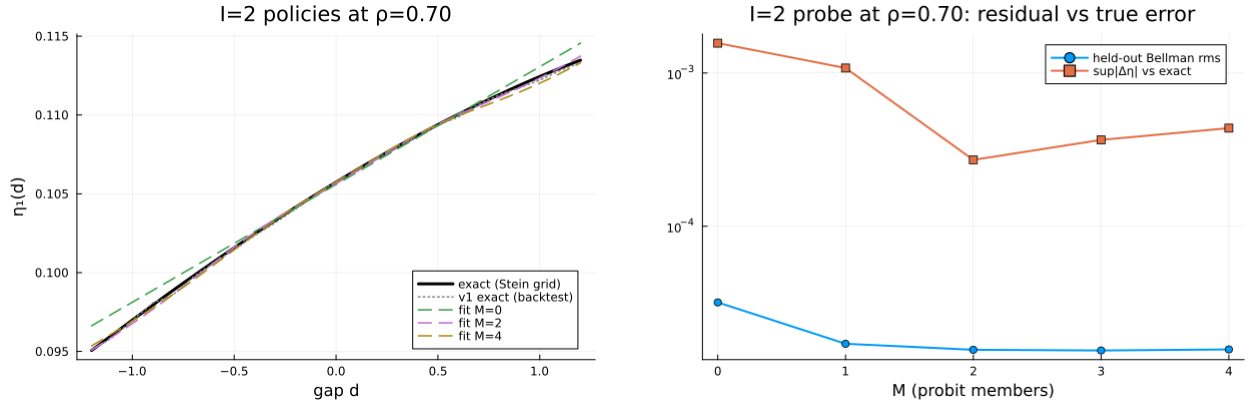
TABLE A1: ACCURACY OF THE CRITIC FAMILY AGAINST EXACT SOLUTIONS

	exact benchmark	out-of-sample rms residual	$\sup \Delta\eta $ vs. exact
$I = 2$	one-dimensional solve	1.6×10^{-5}	2.8×10^{-4}
$I = 3$	two-dimensional solve	1.7×10^{-5}	5.9×10^{-4}
$I = 4$	—	3.0×10^{-5}	—
$I = 5$	—	4.7×10^{-5}	—

Notes: Test game $\sigma = 3.9$, $\rho_a = 0.87$, $\kappa = 1.1$, $\nu_a = 0.1$, discount anchored at $\beta = 0.70$ where the exact solvers converge; orbit-tied pairwise-gap probit basis; second-order conditions verified at every row. No exact solver exists beyond $I = 3$; there the out-of-sample residual and the stationarity certificate provide the audit.

A.3 Two-Firm Case: Proofs

Numerical statements in Section 3.5 and in this appendix use the fitted equilibrium at the illustrative parameters $\sigma = 3.9$, $\rho_a = 0.87$, $\kappa = 1.1$, $\beta = 0.95$, $\nu_a = 0.1$, $\nu_b = 0$,

FIGURE A1: THE $I = 2$ AUDIT: POLICIES AND THE RESIDUAL LADDER


Notes: Left: fitted marketing policies at $M = 0, 2, 4$ probit members against the exact solutions at $I = 2$. Right: the out-of-sample Bellman residual and the true policy error $\sup |\Delta\eta|$ against the exact solution as members are added; both flatten after $M = 2$, and the residual tracks the true error. Test-game parameters as in Table A1. Source: model computations.

simulated over $T = 200,000$ periods; the comparative statics in Figures 1 and 2 state their own parameters in the notes.

Logistic shares (20). The fixed point implies $\ln \frac{s_1}{s_2} - (\sigma - 1) \ln \frac{1-s_1}{1-s_2} = \zeta$; with $s_2 = 1 - s_1$ the left side is $\sigma \ln \frac{s_1}{1-s_1}$. Invert and differentiate; $s'' = s'(1 - 2s)/\sigma$ vanishes at the tie.

Proof of Proposition 1. By (20), $\zeta \mapsto s$ is a strictly increasing bijection, so maximize $s(1-s)[1 + 2(\tilde{\sigma} - 1)s]$ over s . Its derivative, $1 - 2s + 4(\tilde{\sigma} - 1)s - 6(\tilde{\sigma} - 1)s^2$, is a concave parabola, positive at $s = \frac{1}{2}$ (value $\frac{\tilde{\sigma}-1}{2}$) and negative at $s = 1$ (value $-1 - 2(\tilde{\sigma} - 1)$): exactly one root in $(\frac{1}{2}, 1)$, the stated s^* . Globally, the reflection identity $h(\zeta) - h(-\zeta) = s(1-s)[\pi'(s) - \pi'(1-s)]/\sigma$ keeps the leader branch above the laggard branch pointwise when $\pi'' > 0$. In the test game of Appendix A.2, where $\omega = 1$ and hence $\tilde{\sigma} = \sigma = 3.9$, the peak share is $s^* = 0.64$, and the fitted equilibrium policy peaks at the larger lead $\zeta \approx 4.7$. The incentive peak is a property of the policy profile rather than the ergodic distribution: the gap drifts by $(\rho_a - 1)\zeta + \Delta\eta(\zeta)$, and at the baseline parameters the fitted policy slope never exceeds 0.03 while depreciation pulls at $1 - \rho_a = 0.13$, so the drift is strictly negative for $\zeta > 0$ and the stationary law is unimodal at the tie. An interior mode requires $\Delta\eta(\bar{\zeta}) = (1 - \rho_a)\bar{\zeta}$ with the policy crossing from above, the threshold displayed in Figure 1.

Proof of Proposition 2. To first order in β , the first-order condition gives $\kappa\eta_1(\zeta) = \beta \mathbb{E}[h(\rho_a\zeta + \varepsilon_\zeta)] + O(\beta^2)$. Differencing across the two firms, differentiating at the tie,

and applying Stein's lemma together with the reflection $s(-\zeta) = 1 - s(\zeta)$ yields the displayed formula, with $h(\varepsilon) - h(-\varepsilon) = \frac{2(\tilde{\sigma}-1)}{\sigma\tilde{\sigma}}s(1-s)(2s-1) \geq 0$ and equality exactly when π is linear in s . Locally, $h'(0) = s'(0)^2\pi''(\bar{s})$ by the logistic lemma: the feedback is the shock-weighted asymmetry of the flow incentive. Fitted equilibria put γ at 0.017 at $\beta = 0.70$ and 0.024 at $\beta = 0.95$ (closed-loop persistence $\rho_a + \gamma = 0.894$, variance amplification $(1 - \rho_a^2)/(1 - (\rho_a + \gamma)^2) = 1.22$).

Atomistic benchmark. Let a firm be atomistic under a Kimball demand with local elasticity $\sigma(x)$ at relative size x , constant marginal cost, the state entering as an effective price pe^{-z} , and the price set optimally. The envelope theorem gives $d \ln \pi / dz = \sigma(x)$ exactly, hence

$$\frac{d^2 \ln \pi}{dz^2} = \sigma'(x) \cdot \frac{d \ln x}{dz}, \quad \text{where} \quad \frac{d \ln x}{dz} = \frac{\sigma(\sigma-1)}{\sigma-1+\varepsilon_\mu} > 0,$$

with ε_μ the superelasticity. An atomistic markup varies only through $\sigma'(x)$, so the proportional return to the state declines in size if and only if markups rise with size: the atomistic analogue of γ is nonpositive whenever markups are (weakly) increasing in size. Amplification requires the wedge between the perceived and the physical elasticity that only internalization creates; under Cournot-CES the physical elasticity is constant while the margin rises with the share.

Proof of Proposition 3 and the closed forms (21). The wage bill is $w\ell_i = \frac{\tilde{\sigma}-1}{\tilde{\sigma}}s_i(1-s_i)$ (Appendix A.1); $s_2 = 1 - s_1$ makes $s_1(1-s_1) = s_2(1-s_2)$ an identity, so $\ell_1 = \ell_2$ at every state. (Bertrand instead gives $\ell_1/\ell_2 = (\sigma s_1 + s_2)/(\sigma s_2 + s_1) > 1$; at $I \geq 3$ the map $\ell_i \propto s_i(1-s_i)$ is humped and differs from every efficient allocation generically.) For the indices, the planner's problem is strictly concave with $\ell_i^* \propto e^{\hat{z}_i}$, the equilibrium sets $\ell_i = L/2$, and the common constants cancel in the ratios, leaving the cosh expressions in (21). The exact facts follow by differentiation: $(\ln \mathcal{M})'(\zeta) = [\tanh(\zeta/2) - \tanh(\zeta/2\sigma)]/[2(\sigma-1)] > 0$ for $\zeta > 0$; only Z^* depends on the source of the gap, and at general I it equals the customer-capital-weighted mean of productivity; and $\ln Z^{\text{eq}} = \ln Z^* - \ln \mathcal{M}$. Corollary 1 is the partial-derivative computation, and the no-customer-capital statement follows from

$$\frac{d \ln Z^*}{d\alpha} - \frac{d \ln \mathcal{M}}{d\alpha} = \frac{\tanh(\alpha/2\sigma)}{2(\sigma-1)} > 0 \quad \forall \alpha > 0,$$

so the frontier gain dominates the markup wedge pointwise and $Z^{\text{eq}} \geq 1$ whenever $\beta_b = 0$.

Ergodic joint distribution. In gap coordinates the dynamics are exact: $\alpha' = \rho_a \alpha + \varepsilon_\alpha$, $\zeta' = \rho_a \zeta + \Delta\eta(\zeta) + \varepsilon_\alpha + \varepsilon_\beta$, $\beta_b = \zeta - \alpha$; with $v_b = 0$ the demand gap has the representation $\beta_{b,t} = \sum_{k \geq 0} \rho_a^k \Delta\eta(\zeta_{t-1-k})$, accumulated past marketing. Replacing $\Delta\eta$ by its linearization $\gamma\zeta$ yields a Gaussian system with closed-loop persistence $\rho_\zeta = \rho_a + \gamma$ and stationary moments

$$\begin{aligned} \sigma_\alpha^2 &= \frac{2v_a^2}{1 - \rho_a^2}, & \sigma_{\alpha\zeta} &= \frac{2v_a^2}{1 - \rho_a\rho_\zeta}, \\ \sigma_\zeta^2 &= \frac{2(v_a^2 + v_b^2)}{1 - \rho_\zeta^2}, & \sigma_{\alpha\beta_b} &= \frac{2v_a^2 \rho_a \gamma}{(1 - \rho_a\rho_\zeta)(1 - \rho_a^2)}, \end{aligned} \quad (\text{A8})$$

so the sign of the alignment covariance is the sign of γ : zero under constant markups, positive here. Simulation of the fitted nonlinear policy ($T = 200,000$) matches the closed forms, with $\text{sd}(\beta_b)$ 0.047 against 0.044, $\text{corr}(\alpha, \beta_b)$ 0.618 against 0.620, and $\text{corr}(\alpha, \zeta)$ 0.993 against 0.994; the linearization is tight because the fitted policy is near-linear on the ergodic support. The entrenchment region follows from the closed forms: $Z^* < 1$ if and only if $\alpha(\alpha + 2\beta_b) < 0$. Simulated frequencies, with the Gaussian arcsine values $\text{Pr} = \frac{1}{2} - \arcsin(\text{corr})/\pi$ in parentheses:

$$\text{Pr}(\alpha\beta_b < 0) = 0.288 \text{ (0.287)}, \quad \text{Pr}(\text{sign } \zeta \neq \text{sign } \alpha) = 0.037 \text{ (0.035)}.$$

Evaluating the exact indices on the simulated distribution gives $\mathbb{E}[\ln Z^*] = 0.0042$, $\mathbb{E}[\ln \mathcal{M}] = 0.0031$, $\mathbb{E}[\ln Z^{\text{eq}}] = 0.0010$, against 0.0009 in the no-marketing benchmark on the same shocks: a net allocative gain of 1.1×10^{-4} , while the ergodic marketing bill absorbs roughly 13% of revenue. To second order, with each term reproducing its simulated value within 2%,

$$\mathbb{E}[\ln Z^{\text{eq}}] \approx \frac{\sigma_\alpha^2 + 2\sigma_{\alpha\beta_b}}{8\sigma(\sigma - 1)} - \frac{\sigma_{\beta_b}^2}{8\sigma} : \quad (\text{A9})$$

productivity dispersion helps, damped by $\sigma(\sigma - 1)$ because markups withhold most of the gain; alignment helps with the same coefficient; demand-gap variance is pure loss.

A.4 Welfare, the Planner, and the Hsieh–Klenow Bridge

Derivation of Lemma 1. Household optimality (Appendix A.8) fixes the wage and each industry's expenditure at one and prices profits at β , so aggregate welfare is $\frac{1}{1-\beta} \mathbb{E}_g[W]$ with $W = \ln C^{\text{eq}} - L^{\text{prod}} - L^{\text{mkt}}$ per industry; the closure requires log consumption utility and linear labor disutility, since with curvature the labor level would respond and the accounting would carry an additional margin. Four steps deliver the decomposition.

Labor pools. Markup pricing sets price over average cost at $\tilde{\mu}_i$, so firm i 's wage bill is $\ell_i = s_i / \tilde{\mu}_i$ and $L^{\text{prod}} = \sum_i s_i / \tilde{\mu}_i = 1 / \bar{\mu}$; L^{mkt} is the sum of the marketing bills.

Output and frontier. Substituting ℓ_i into the aggregator (15) gives the closed form (A3). Maximizing (15) over (ℓ_i) at fixed total labor L gives $\ell_i^* \propto (a_i b_i^v)^{\tilde{\sigma}/\sigma}$ and $C^* = Z^* L^\omega$, with Z^* as in the lemma.

Dispersion wedge. Define $Z^{\text{eq}} \equiv C^{\text{eq}} / (L^{\text{prod}})^\omega$ and $\mathcal{M} \equiv Z^* / Z^{\text{eq}}$. The share fixed point (13) gives $\ell_i \propto \ell_i^* \tilde{\mu}_i^{-\tilde{\sigma}}$: the equilibrium distorts the efficient allocation by a power of the markup alone. Substituting the relative allocation ℓ_i / ℓ_i^* into (15),

$$\ln \mathcal{M} = \frac{\sigma}{\sigma-1} \left[\frac{\omega(\sigma-1)}{\sigma} \ln \mathbb{E}^* [\tilde{\mu}^{-\tilde{\sigma}}] - \ln \mathbb{E}^* [\tilde{\mu}^{-\tilde{\sigma}\omega(\sigma-1)/\sigma}] \right],$$

with $\mathbb{E}^*$ the expectation under the frontier weights $\propto (a_i b_i^v)^{\tilde{\sigma}/\sigma}$. The fixed point also gives $(a_i b_i^v)^{\tilde{\sigma}/\sigma} \propto s_i \tilde{\mu}_i^{\tilde{\sigma}-1}$; substituting, the second exponent is zero by (12), the first moment is $\sum_i s_i / \tilde{\mu}_i = 1 / \bar{\mu}$ up to the common normalizer, and $\tilde{\sigma} - 1 = \omega(\sigma-1)\tilde{\sigma}/\sigma$ collapses the expression to the lemma's

$$\mathcal{M} = \left[\frac{(\sum_i s_i \tilde{\mu}_i^{\tilde{\sigma}-1})^{\frac{1}{\tilde{\sigma}-1}}}{\bar{\mu}} \right]^\omega \geq 1,$$

with equality if and only if the $\tilde{\mu}_i$ are equal, by power-mean monotonicity ($\tilde{\sigma} - 1 > -1$). At $\omega = 1$, where $\tilde{\sigma} = \sigma$, this is the standard CES misallocation formula.

Level wedge and assembly. The two definitions give $\ln C^{\text{eq}} = \ln Z^* - \ln \mathcal{M} + \omega \ln L^{\text{prod}}$, so

$$W = \ln C^{\text{eq}} - L^{\text{prod}} - L^{\text{mkt}} = \ln Z^* - \ln \mathcal{M} - \Lambda - L^{\text{mkt}}, \quad \Lambda \equiv L^{\text{prod}} - \omega \ln L^{\text{prod}} = \frac{1}{\bar{\mu}} + \omega \ln \bar{\mu},$$

which is (23). The map $L \mapsto L - \omega \ln L$ is strictly convex with minimum $\omega - \omega \ln \omega$ at $L = \omega$, so $\Lambda \geq \omega - \omega \ln \omega > 0$ with equality if and only if $\bar{\mu} = 1/\omega$. The first-best flow sets $\ell_i = \ell_i^*$ and $L = \omega$: $\ln \mathcal{M} = 0$ and Λ at its minimum. $\square$

Zero-sum social value of customer capital. At every state the allocative part of the flow, $W + L^{\text{mkt}} = \ln Z^* - \ln \mathcal{M} - \Lambda$, has $\sum_i \partial(W + L^{\text{mkt}})/\partial b_i = 0$: a uniform scaling $\mathbf{b} \rightarrow c \mathbf{b}$ leaves the weights $b_i^\nu / \sum_k b_k^\nu$ unchanged, hence shares, labor pools, C^{eq} , and Z^* unchanged (the normalization in (4) is chosen to deliver this property), so W is constant along the shift; differentiate at $c = 0$. Because the shadow-value recursion below is linear with common decay, the zero-sum property extends to the discounted shadow values. This has two consequences: with unrestricted marketing the planner's policies sum to zero across firms, and at any symmetric state the planner does not market at all, while each firm's private flow value there is $\phi_{ii} = \pi'(\bar{s}) \chi(\bar{s}) (I - 1)/I = 0.0434$ at the calibration ($I = 5$) and the social value is zero to machine precision, so marketing at the modal symmetric state is entirely business stealing.

Planner's first-order condition. The planner controls all policies simultaneously, so the envelope theorem removes the cross-policy terms that generate the equilibrium's strategic feedback, leaving the recursion (35), which is the equilibrium's Stein problem with the private flow ϕ_{ii} replaced by the social flow g_i and without the feedback block, and is therefore easier to compute. In the first-best regime the flow is (37), obtained by differentiating $\ln Z^*$, with the productivity gradient $\partial \ln Z^* / \partial a_i = e^{\hat{z}_i} / [(\sigma - 1) \sum_j e^{\hat{z}_j}] > 0$. In the constrained (Cournot) regime, using the share Jacobian (A1),

$$g_i^C = \frac{1}{\sigma - 1} \left[\frac{\partial \ln N}{\partial z_i} - \frac{b_i^\nu}{\sum_k b_k^\nu} \right] + \frac{2(\sigma - 1)}{\sigma} \left[s_i \chi_i - \hat{\chi}_i \sum_j s_j \chi_j \right], \quad (\text{A10})$$

$$\frac{\partial \ln N}{\partial z_i} = s_i - (\sigma - 1) \left[\frac{s_i \chi_i}{1 - s_i} - \hat{\chi}_i \sum_j \frac{s_j \chi_j}{1 - s_j} \right],$$

where the second bracket is the markup-withholding term. Tilting the state toward large firms lowers production-labor demand, which raises flow utility through lower labor disutility; this gives the planner a mild preference for concentration that does not appear in the log index $\ln Z^{\text{eq}}$. All closed forms are verified against central finite differences (maximum gradient error 2.5×10^{-10} ; zero-sum residual 10^{-14}). Since $\eta \equiv 0$ is feasible for the planner, the planner's value weakly exceeds the no-marketing benchmark in both regimes; the strict gain comes from aligning customer capital with productivity.

Derivation of Lemma 2. In the subsidized game firm i 's critic and actor are

$$\tilde{V}_i(\mathbf{x}) = \pi_i(\mathbf{z}) + \tau_i(\mathbf{x}) \eta_i(\mathbf{x}) - K_i(\eta_i(\mathbf{x}), \mathbf{b}) + \beta \mathbb{E}[\tilde{V}_i(\mathbf{x}') | \mathbf{x}], \quad (\text{A11})$$

$$K'_i(\eta_i(\mathbf{x}), \mathbf{b}) = \tau_i(\mathbf{x}) + \frac{\beta}{\nu_b^2} \mathbb{E}[\varepsilon_i^b \tilde{V}_i(\mathbf{x}') | \mathbf{x}]. \quad (\text{A12})$$

Fix the candidate policies at η^P and define the pair $(\tau, \tilde{\mathbf{V}})$ jointly: $\tilde{V}_i$ as the value of (A11) under the planner's policies, and

$$\tau_i(\mathbf{x}) \equiv K'_i(\eta_i^P(\mathbf{x}), \mathbf{b}) - \frac{\beta}{\nu_b^2} \mathbb{E}^P[\varepsilon_i^b \tilde{V}_i(\mathbf{x}') | \mathbf{x}]. \quad (\text{A13})$$

Given τ the values solve a discounted linear recursion; given the values τ is explicit; the composite map is a contraction at discount β , so the pair exists and is unique. By construction the subsidized first-order condition (A12) holds at η^P in every state, and the second-order condition is unchanged because the subsidy is linear in η_i and the curvature K''_i is untouched, so the planner's policies form a marketing equilibrium of the subsidized game. Substituting the planner's actor (38) into (A13) gives $\tau_i = \frac{\beta}{\nu_b^2} \mathbb{E}^P[\varepsilon_i^b (V^P - \tilde{V}_i)(\mathbf{x}') | \mathbf{x}]$, and adding and subtracting the subsidized total $\tilde{V}^\Sigma \equiv \sum_k \tilde{V}_k$ delivers the two-wedge form (41). Lump-sum financing through the household budget leaves (11) and the statics unchanged.

Hsieh–Klenow mapping. The physical customer-capital stock is the level b_i of (7), the demand shifter is b_i^ν , and $\bar{\alpha}_B \equiv \alpha_B / (1 - \alpha_L) = \frac{\tilde{\sigma}}{\sigma} \nu \in (0, 1)$ is the *total* revenue elasticity of customer capital of Section 4.1, the elasticity the concentrated regression recovers. In the Cournot equilibrium $R_i = s_i$ and, from the share fixed point,

$$\ln R_i = \bar{\alpha}_B \ln b_i + \ln \tilde{a}_i + \text{const}, \quad \text{with} \quad \ln \tilde{a}_i \equiv \frac{\tilde{\sigma}}{\sigma} \ln a_i + (\tilde{\sigma} - 1) \ln(1 - s_i), \quad (\text{A14})$$

which is (27), verified in the simulated equilibrium to 10^{-15} . The estimating equation of Section 4.1 therefore holds exactly in the model, with three implications: the revenue-productivity index $\tilde{a}$ conflates the structural index $\hat{a}_i = a_i^{\tilde{\sigma}/\sigma}$ with the oligopoly markup (under monopolistic competition $\tilde{a}_i = \hat{a}_i$ exactly); the labor revenue product equals the markup, $R_i/\ell_i = \mu_i$; and the customer-capital revenue product is $\ln(R_i/b_i) = \ln \tilde{a}_i - (1 - \bar{\alpha}_B) \ln b_i + \text{const}$, which is the object whose dispersion the data measure.

Free- b planner and the two-wedge decomposition. Let the planner reallocate the aggregate stock freely, $\sum_i b_i = \bar{b}$, with labor allocated efficiently at the resulting state. From the frontier of Lemma 1, measured TFP per unit of aggregate stock is

$$A(\mathbf{a}, \mathbf{b}) = \left(\sum_i (a_i b_i^\nu)^{\tilde{\sigma}/\sigma} \right)^{\frac{\sigma}{\tilde{\sigma}(\sigma-1)}} / \left(\sum_i b_i \right)^{\frac{\nu}{\sigma-1}}.$$

The stock enters the maximand with exponent $\nu\tilde{\sigma}/\sigma = \bar{\alpha}_B < 1$, so the program is strictly concave with a unique interior optimum,

$$b_i^* \propto a_i^{\frac{\tilde{\sigma}}{\sigma(1-\bar{\alpha}_B)}} \iff \ln b_i^* = \frac{\tilde{\sigma}}{\sigma(1-\bar{\alpha}_B)} \ln a_i + \text{const}, \quad (\text{A15})$$

perfect alignment; at the constant-returns closure $(\sigma/\tilde{\sigma})(1-\bar{\alpha}_B) = 1$, so the efficient line is $\ln b_i^* = \ln a_i$. Substituting back, the optimized TFP is the Hsieh–Klenow power mean at curvature $(\sigma/\tilde{\sigma})(1-\bar{\alpha}_B)$,

$$A^* = \left[\sum_i a_i^{\frac{\tilde{\sigma}}{\sigma(1-\bar{\alpha}_B)}} \right]^{\frac{\sigma(1-\bar{\alpha}_B)}{\tilde{\sigma}(\sigma-1)}},$$

independent of $\bar{b}$ and of the equilibrium stocks; at $\omega = 1$ ($\tilde{\sigma} = \sigma$) it reduces to the familiar $[\sum_i a_i^{1/(1-\bar{\alpha}_B)}]^{(1-\bar{\alpha}_B)/(\sigma-1)}$ form. Letting A^L denote measured TFP with labor reallocated efficiently but $\mathbf{b}$ held fixed, the static loss splits exactly into two nonnegative wedges,

$$\ln \frac{A^*}{A^{\text{eq}}} = \underbrace{\ln \mathcal{M}}_{\text{markup wedge}} + \underbrace{\ln \frac{A^*}{A^L}}_{\text{demand misallocation}}, \quad (\text{A16})$$

where $\ln \mathcal{M} = \ln A^L - \ln A^{\text{eq}}$ coincides with the markup wedge of Lemma 1 (the weight normalizations cancel). For $(\ln a_i, \ln b_i)$ jointly normal and demeaned within the market, expanding both power means to second order (exact as $I \rightarrow \infty$ under log-normality),

$$\ln \frac{A^*}{A^L} = \frac{\bar{\alpha}_B (1 - 1/I) \tilde{\sigma}}{2\sigma(1-\bar{\alpha}_B)(\sigma-1)} \text{Var} \left(\ln a_i - \frac{\sigma}{\tilde{\sigma}} (1 - \bar{\alpha}_B) \ln b_i \right), \quad (\text{A17})$$

the variance of the distance from the free- b line (A15), which by (A14) is the variance of the log revenue product of customer capital up to the markup term, the empirical object of Section 4.1; relative to the $\omega = 1$ form the coefficient carries the extra factor $\tilde{\sigma}/\sigma$. The quadratic form links the wedge to the measured dispersion; every wedge we report from the model is computed exactly from the power means. This calculation is in output units,

normalized by $(\sum_i b_i)^\nu$, while welfare weights customer capital through $\sum_i b_i^\nu$; the bridge between the two is $[\nu \ln \sum_i b_i - \ln \sum_i b_i^\nu] / (\sigma - 1)$ evaluated at $\mathbf{b}^*$ versus the equilibrium stocks. In the calibrated panel the equilibrium distribution is more dispersed than $\mathbf{b}^*$, the bridge term is positive, and the Hsieh–Klenow loss is a lower bound on the welfare-unit static loss.

A.5 Static Accounting: Flexible-Input Form

Revenue after substituting labor out. Demand is CES with elasticity σ and the weights of (4); production uses labor alone, and the stock enters through the demand weight. Folding the weight into effective output, with ω the labor exponent of (5) and $\tilde{\sigma}$ the rescaled elasticity of (12), labor’s first-order condition is $w\ell_i = (\omega/\mu)R_i$ in every period. Substituting it out and collecting terms gives the share fixed point of Section 4.1, with

$$\bar{\alpha}_B \equiv \frac{\alpha_B}{1 - \alpha_L} = \frac{\tilde{\sigma}}{\sigma} \nu, \quad \hat{a}_i = a_i^{\tilde{\sigma}/\sigma}, \quad (\text{A18})$$

where ν translates the accumulated stock into the demand weight. The regression recovers $\bar{\alpha}_B$ whatever ω and ν separately are, since they enter only through the ratio.

Why only the elasticity enters the loss. Marginal and average revenue products of the stock are proportional under (26), so equating the average product is efficient and the optimum is (A15). Writing $\text{arp}_i = \log \text{ARPB}_i$, aggregate output at the observed allocation is $\bar{b} \mathbb{E}_b[e^{\text{arp}}]$ under stock-share weights and $\bar{b}(\mathbb{E}_b[e^{\text{arp}/(1-\bar{\alpha}_B)}])^{1-\bar{\alpha}_B}$ at the optimum, which gives (28) for arp normal under those weights. Centering and variance are both taken under stock-share weights, as the derivation requires; a second moment about the equal-weighted mean is a different object and overstates it. Nothing beyond $\bar{\alpha}_B$ is calibrated, so the demand elasticity, the labor elasticity, and the translation ν all drop out, leaving $\bar{\alpha}_B$ as the only quantity the number rests on.

Variable markups. Under Cournot the markup varies with the share and the same $(1 - s_i)$ multiplies every input’s first-order condition, so labor becomes $\ell_i \propto (1 - s_i)R_i$ and the share fixed point carries a markup term $(1 - s_i)^{\tilde{\sigma}-1}$. The regression slope is then the total elasticity rather than the structural one, separating the two by the slope of $\log(1 - s_i)$ on the log stock, -0.0209 (0.0048). Every object on the technology side moves and the gain does not, because reallocating the stock moves shares as well and (28) calls for the total elasticity, which is 0.598 either way. We leave the markup out of the baseline

for that reason and because the adjustment requires the mean of $1 - s_i$ on public firms alone, where the weighting is unsettled.

Evidence on α_L . The flexible-input cost share puts the labor exponent near one rather than near the closure. Reading it off the flexible input's first-order condition as [Hsieh and Klenow \(2009\)](#) read α and [David and Venkateswaran \(2019\)](#) read their elasticity, $\omega = \mu(s_i) \times \text{flexible cost}_i / R_i$, with $\mathbb{E}_{sw}[1/\mu] = 0.610$ at $\sigma = 4$ on Compustat 2014–2022 (29,030 firm-years, sales-weighted):

Flexible-input measure	share of R	$\omega, \sigma = 4$	$\omega, \sigma = 8$
Operating expense	0.831	1.361	1.166
Operating expense net of marketing and R&D	0.775	1.270	1.089
Cost of goods sold	0.666	1.091	0.935

Cost of goods sold is the flexible input, the broader measures carrying overhead, research, and marketing. Regressing revenue on the stock and labor agrees: the estimated labor revenue elasticity is 0.760 (0.025), against the $\alpha_L = 0.378$ the closure imposes and the 0.750 that $\omega = 1$ predicts, so $\omega = 1.013$ (0.033). Both routes sit far from the 0.504 that constant returns in the bundle require at $\sigma = 4$ (Table 2).

Fixed cost inside cost of goods sold would reconcile them, since it inflates the measured share. With flexible cost equal to cost of goods sold net of a fixed component F , the requirement is $F/R = 0.666 - 0.610\omega$: eight percent of cost of goods sold being overhead takes 1.09 to 1.00, while the constant-returns closure at 0.504 needs 54% of it to be fixed. The question is measurable from a cost decomposition or from the cyclicity of cost of goods sold with output, and we have not settled it.

Placement of the stock when it is not a factor. Counting the stock inside productivity removes the expression of its misallocation as revenue-TFP dispersion, not the misallocation itself. With labor allocated efficiently, consumption is labor times a weighted mean of productivity,

$$\mathcal{C} = \mathcal{L} \left[\sum_i \theta_i a_i \right]^{\frac{1}{\sigma-1}}, \quad \theta_i = \frac{b_i^\nu}{\sum_j b_j^\nu}, \quad (\text{A19})$$

so the planner wants weight on productive firms and placement of the stock stays first order. The static optimum in (A19) is a corner, since θ is a probability distribution that can be concentrated arbitrarily and a flat marginal product removes the penalty for piling production into one firm; an interior optimum requires convex accumulation

cost, which is dynamic. A resource-neutral benchmark avoids the corner by holding the multiset $\{b_i\}$ fixed and reassigning it across firms, for which the optimum is assortative. Sales-weighted across markets with at least five firms, the best permutation raises output by 18.4%, the corner by 39.8%, and an assignment independent of productivity lowers it by 32.9%, so the observed assignment already places customer capital with the more productive firms, imperfectly.

What limits the alignment estimates. The revenue-productivity index is the residual of the regression that sets the translation between the stock and the demand weight, so $\text{Cov}(\log \tilde{a}, \log b) = 0$ by construction in the estimating dimension and the measured alignment is chosen rather than observed. A nonzero covariance between $\log \tilde{a}$ and $\log b$ arises only when the elasticity used to form $\tilde{a}$ differs from that panel's own slope, 0.987, and the benchmarks of the previous paragraph move with the same choice. Since $R_i/\ell_i = \mu(s_i)w$ carries no productivity, no independent measure of a exists to check the residual against, and the circularity is structural rather than a small-sample matter. An instrument for the translation, or observed prices, would make the alignment an estimate; the scanner data of Appendix C are the natural source and we have not yet used them for this.

A.6 Rational Inattention and Customer Base

With multiple firms, a rational inattention framework, adapted from Wu (2024), maps the consumer's choice across firms' product bundles to customer capital. Households have limited attention to allocate across competing options: each household h chooses the probability q_{hi} of purchasing from firm i by solving

$$\max_{q_{hi}, c_{hi}} \sum_i q_{hi} \log c_{hi} - \psi \sum_i q_{hi} \log \frac{q_{hi}}{\bar{q}_i},$$

where c_{hi} is consumption and

$$\bar{q}_i = \frac{b_i^\nu}{\sum_j b_j^\nu}$$

is the default attention allocation based on customer capital b_i . The choice respects the adding-up constraint $\sum_i q_{hi} = 1$ and the unit-budget constraint $c_{hi} = \frac{1}{p_i}$.

The solution is the expenditure share

$$q_i = \frac{b_i^\nu c_i^{1/\psi}}{\sum_j b_j^\nu c_j^{1/\psi}} = \frac{b_i^\nu p_i^{-1/\psi}}{\sum_j b_j^\nu p_j^{-1/\psi}},$$

where ψ is the cost of directing choice toward the more productive firms. The formulation generates expected utility $\log \left(\frac{\sum_i b_i^\nu c_i^{1/\psi}}{\sum_i b_i^\nu} \right)^\psi$, which is equivalent to the baseline CES utility when $\psi = \frac{\sigma}{\sigma-1}$ and the attention default is read at the aggregator's exponent, $\bar{q}_i \propto b_i^{\nu/\sigma}$, so that the ν of this appendix is the ν/σ of (4), yielding

$$\log \left(\frac{\sum_i b_i^\nu c_i^{\frac{\sigma-1}{\sigma}}}{\sum_i b_i^\nu} \right)^{\frac{\sigma}{\sigma-1}}.$$

The model further relates consumption to firm productivity through $c_{it} = a_{it}^{1/(\sigma-1)} \ell_{it}^\omega$ as in (5), where a_{it} is firm-specific productivity and ℓ_{it} is labor input.

Customer capital b_i is the stock of marketing and brand-building investment that shifts the default attention allocation $\bar{q}_i$ in the firm's favor, lowering consumers' cognitive cost of choosing its products. Marketing therefore pays even when it leaves product quality and characteristics unchanged: it operates on the attention allocation of boundedly rational consumers. Firms with larger customer capital maintain higher market shares and can charge premium prices, a value of brand recognition beyond productivity, in line with a growing literature on the centrality of customers in market share.

A.7 Policy Extensions

We first consider a subsidy for a policymaker who observes firms' demand and productivity differences separately. The optimal policy that corrects the markup distortions removes firms' revenue incentives and reimburses each firm with the full social surplus it creates, through the gross transfer

$$T(q) = \frac{\sigma}{\sigma-1} \frac{b_i^{\nu/\sigma} q^{\frac{\sigma-1}{\sigma}}}{\sum_j a_j b_j^\nu} - \frac{b_i^{\nu/\sigma} q^{\frac{\sigma-1}{\sigma}}}{\sum_{j \neq i} a_j b_j^\nu + b_i^{\nu/\sigma} q^{\frac{\sigma-1}{\sigma}}}.$$

With the other firms all pricing at marginal cost, the post-transfer problem of the focal firm i is

$$\max_q \frac{\sigma}{\sigma-1} \frac{b_i^{\nu/\sigma} q^{\frac{\sigma-1}{\sigma}}}{\sum_j a_j b_j^\nu} - q a_i^{1/(1-\sigma)}.$$

The optimal choice of quantity is $q_i^* = \frac{a_i b_i^{\frac{\nu\sigma}{\sigma-1}}}{\sum_j a_j b_j^\nu}$ and the price is the marginal cost $p_i^* = a_i^{1/(1-\sigma)}$. This implements the static optimal allocation. In the subsidized equilibrium, every firm receives the share of surplus it creates:

$$T_i^* = \frac{1}{\sigma-1} \frac{a_i b_i^\nu}{\sum_j a_j b_j^\nu}.$$

We again linearize the post-transfer payoff around the equal-demand state $(\alpha, \mathbf{0})$:

$$T_j^* \approx \frac{1}{\sigma-1} \bar{s}_i^*(\alpha) \left(\nu \ln b_i - \sum_j \bar{s}_j^*(\alpha) \nu \ln b_j \right) + \frac{1}{\sigma-1} \frac{a_i}{\sum_j a_j}.$$

λ -subsidy: partial markup offset from size alone. When the policymaker observes only the composite z_i , the implementable instrument of Section 5 is a size-based output subsidy that offsets the fraction λ of each firm's log markup, $\mu_i \rightarrow \mu_i^{1-\lambda}$. Prices $p_i = \mu_i^{1-\lambda} a_i^{-1/(\sigma-1)}$ give the share fixed point

$$s_i^\lambda \propto e^{z_i} (1 - s_i^\lambda)^{(1-\lambda)(\tilde{\sigma}-1)}, \quad (\text{A20})$$

the Cournot system at the pseudo-elasticity $\tilde{\sigma}_\lambda = 1 + (1-\lambda)(\tilde{\sigma}-1)$: $\lambda = 0$ reproduces Cournot and $\lambda = 1$ is marginal-cost pricing, the first-best statics given $\mathbf{z}$ (softmax shares). The static aggregates are closed form,

$$\begin{aligned} \ln C_\lambda &= (1-\lambda) \ln \frac{\sigma-1}{\sigma} + \frac{\ln N_\lambda}{\sigma-1}, & N_\lambda &= \sum_j e^{z_j} (1 - s_j^\lambda)^{(1-\lambda)(\sigma-1)} / \sum_j b_j^\nu, \\ L_\lambda^{\text{prod}} &= \sum_i s_i^\lambda \left[\frac{(\tilde{\sigma}-1)(1-s_i^\lambda)}{\tilde{\sigma}} \right]^{1-\lambda}, \end{aligned}$$

and both endpoints are certified against the Cournot statics and the softmax limit to machine precision. The subsidy schedule leaves firms the Cournot rent shape at the subsidized shares (rent-preserving incidence), $\pi_i^\lambda = s_i^\lambda / \tilde{\sigma} + \frac{\tilde{\sigma}-1}{\tilde{\sigma}} (s_i^\lambda)^2$, so the dynamic game is the marketing equilibrium of Section 3.4 under the reshaped prize, solved by the method of Section 3.6 at the same β as the equilibrium for every λ . The moat-drift

diagnostic is $\mathbb{E}[\Delta\beta_b] = \eta_1 - \eta_{\text{rival}}$ at the fitted policies, with the rivals held symmetric.

A.8 Aggregation and Household Problem

The household makes static consumption and labor decisions and a dynamic savings decision. The household has flow utility $\log \mathcal{C}_t - \mathcal{L}_t$ and holds assets ω_t , with the budget constraint

$$\dot{\omega}_t = r_t \omega_t + \mathcal{L}_t + \Pi_t - \int_0^1 \sum_{i=1}^I p_{ikt} c_{ikt} dk,$$

where aggregate consumption is Cobb–Douglas across product groups, $\log \mathcal{C}_t = \int_0^1 \log C_{kt} dk$, with C_{kt} the industry bundle of (4). The wage is normalized to 1.

First-order conditions. Let Ξ_t denote the multiplier on the flow budget. The consumption condition for variety i in industry k is $\partial \ln C_{kt} / \partial c_{ikt} = \Xi_t p_{ikt}$; multiplying by c_{ikt} and summing within the industry, homogeneity of degree one of the aggregator (4) gives $\sum_i (\partial \ln C_{kt} / \partial c_{ikt}) c_{ikt} = 1$, so industry expenditure is $E_{kt} = 1/\Xi_t$: equal across industries, the Cobb–Douglas constant-share result. The labor condition is $1 = \Xi_t w_t$; with the wage as numeraire, $\Xi_t = 1$ and hence $E_{kt} = 1$, which is (11). The savings condition is the Euler equation $\Xi_t = \beta(1 + r_{t+1})\Xi_{t+1}$, so $1 + r_{t+1} = 1/\beta$ and the representative portfolio prices profit streams at the household’s discount factor. Linear disutility makes labor supply perfectly elastic at the numeraire wage, the aggregate labor condition of Section 3.2: the clearing condition (10) determines $\mathcal{L}_t$ residually, with no feedback into prices. The share Jacobian this consumption problem induces is (A1) in Appendix A.1.

Linear aggregation over non-settling industries. Industries are ex-ante identical, their states are independent, and each industry’s state follows the same ergodic Markov equilibrium with invariant distribution π . By the law of large numbers across the continuum, cross-sectional averages equal ergodic means at every date: $\ln \mathcal{C}_t = \int_0^1 \ln C_{kt} dk = \mathbb{E}_\pi[\ln C(\mathbf{z})]$ and $\mathcal{L}_t = \mathbb{E}_\pi[L^{\text{prod}} + L^{\text{mkt}}]$ almost surely, both constant over time although no industry’s state converges. Aggregate welfare is therefore $\sum_t \beta^t (\ln \mathcal{C}_t - \mathcal{L}_t) = \frac{1}{1-\beta} \mathbb{E}_\pi[W(\mathbf{z})]$ with W the per-industry flow (23), the object every counterfactual in Section 5 differences.

Multi-product extension. For the empirical analysis, we model multi-product firms with constant elasticity of substitution across products and decreasing returns in pro-

duction (as in [Hottman et al., 2016](#)). Firm-level consumption is

$$c_{it} = \left(\sum_{u=1}^{U_i} \varsigma_{iut} d_{iut}^{\frac{\sigma_u-1}{\sigma_u}} \right)^{\frac{\sigma_u-1}{\sigma_u}}, \quad \sigma_u > \sigma$$

where ς_{iut} are product appeals, $\sigma_u > \sigma$ implies constant markups across product lines, and production at the firm-group level exhibits decreasing returns to scale: $c_{ikt} = a_{ikt}^{1/(\sigma-1)} \ell_{ikt}^\omega$ with ω the labor exponent of (5).

B Estimation Appendix

This appendix details the two-step procedure behind the scanner demand system of Appendix C: it recovers the autoregressive parameters $(\rho_{aa}, \rho_{ba}, \rho_{bb})$ and the variance components of that system, and is separate from the Compustat calibration of Section 4.2, which is matched by minimum distance on the autocovariances of measured productivity growth. We first estimate the autoregressive parameters across a bracket of panel estimators, then recover variance components through GMM moment matching.

B.1 Model Structure

Our empirical model decomposes both productivity and customer capital into persistent and transitory components. For firm j at time t :

$$\begin{aligned} a_{jt} &= a_{jt}^p + \tau_{jt}^a + \bar{a}_j \\ b_{jt} &= b_{jt}^p + \tau_{jt}^b + \bar{b}_j \end{aligned}$$

where τ_{jt}^a and τ_{jt}^b are the transitory shocks, independent of the current and past persistent shocks; $\bar{a}_j$ and $\bar{b}_j$ are the firm fixed effects; ε_{jt}^a and ε_{jt}^b are the innovations of the persistent components. Here we take $\rho_{aa}, \rho_{ba}, \rho_{bb}$ as already recovered from Arellano–Bond and recover the variances $(\sigma_{aT}^2, \sigma_{bT}^2, \sigma_{aP}^2, \sigma_{bP}^2)$. We will recover these parameters from the autocovariance function of the first difference. More precisely, we denote

$$\Sigma_k^x = \text{Cov}(x_{jt} - x_{jt-1}, x_{jt-k} - x_{jt-k-1}), \quad x = a, b$$

It is useful to write out the values in terms of past realizations. More precisely, for any $k \geq 1$

$$a_{jt}^P = \rho_{aa}^k a_{jt-k}^P + error$$

where the error is independent of a_{jt-k}^P . We can do the same with b :

$$b_{jt}^P = \rho_{bb}^k b_{jt-k}^P + \left(\sum_{l=0}^{k-1} \rho_{bb}^l \rho_{ba} \rho_{aa}^{k-l-1} \right) a_{jt-k}^P + error$$

For notational simplicity, let us define

$$Q_k = \sum_{l=0}^{k-1} \rho_{bb}^l \rho_{ba} \rho_{aa}^{k-l-1}$$

Thus

$$b_{jt}^P = \rho_{bb}^k b_{jt-k}^P + Q_k a_{jt-k}^P + error$$

Productivity To start, we impose stationarity and denote the stationary variance of a^P as $Var(a^P)$:

$$Var(a^P) = \rho_{aa}^2 Var(a^P) + \sigma_{aP}^2 \implies Var(a^P) = \frac{\sigma_{aP}^2}{1 - \rho_{aa}^2}$$

We start with $k = 0$:

$$\begin{aligned} \Sigma_0^a &= Var(a_{jt} - a_{jt-1}) = Var(a_{jt}^P - a_{jt-1}^P + \tau_{jt}^a - \tau_{jt-1}^a) \\ &= Var(a_{jt}^P - a_{jt-1}^P) + Var(\tau_{jt}^a - \tau_{jt-1}^a) \\ &= Var(\rho_{aa} a_{jt-1}^P + \varepsilon_{jt}^a - a_{jt-1}^P) + 2\sigma_{aT}^2 \\ &= (1 - \rho_{aa})^2 Var(a^P) + \sigma_{aP}^2 + 2\sigma_{aT}^2 \end{aligned}$$

where the first equality uses the definition, the second equality uses the fact that τ^a is independent of the persistent components, the third equality uses the definition of the transitory variance, and the last equality uses the fact that ε_{jt}^a is independent of past values. Plugging in the stationary variance, we have

$$\Sigma_0^a = \frac{2}{1 + \rho_{aa}} \sigma_{aP}^2 + 2\sigma_{aT}^2$$

Now we move on to $k = 1$:

$$\begin{aligned}
 \Sigma_1^a &= \text{Cov} \left(a_{jt}^P - a_{jt-1}^P + \tau_{jt}^a - \tau_{jt-1}^a, a_{jt-1}^P - a_{jt-2}^P + \tau_{jt-1}^a - \tau_{jt-2}^a \right) \\
 &= \text{Cov} \left(a_{jt}^P - a_{jt-1}^P, a_{jt-1}^P - a_{jt-2}^P \right) + \text{Cov} \left(\tau_{jt}^a - \tau_{jt-1}^a, \tau_{jt-1}^a - \tau_{jt-2}^a \right) \\
 &= \text{Cov} \left(a_{jt}^P - a_{jt-1}^P, a_{jt-1}^P - a_{jt-2}^P \right) - \sigma_{aT}^2 \\
 &= \text{Cov} \left(a_{jt}^P, a_{jt-1}^P \right) - \text{Var} \left(a^P \right) - \text{Cov} \left(a_{jt}^P, a_{jt-2}^P \right) + \text{Cov} \left(a_{jt-1}^P, a_{jt-2}^P \right) - \sigma_{aT}^2
 \end{aligned}$$

where the first equality uses the independence of τ^a from the persistent values. Using the sequential form,

$$\text{Cov} \left(a_{jt}^P, a_{jt-1}^P \right) = \text{Cov} \left(a_{jt-1}^P, a_{jt-2}^P \right) = \rho_{aa} \text{Var} (a^P), \quad \text{Cov} \left(a_{jt}^P, a_{jt-2}^P \right) = \rho_{aa}^2 \text{Var} (a^P),$$

and plugging in these values, we have

$$\Sigma_1^a = -\frac{1 - \rho_{aa}}{1 + \rho_{aa}} \sigma_{aP}^2 - \sigma_{aT}^2$$

Now we move on to $k \geq 2$:

$$\begin{aligned}
 \Sigma_k^a &= \text{Cov} \left(a_{jt}^P - a_{jt-1}^P + \tau_{jt}^a - \tau_{jt-1}^a, a_{jt-k}^P - a_{jt-k-1}^P + \tau_{jt-k}^a - \tau_{jt-k-1}^a \right) \\
 &= \text{Cov} \left(a_{jt}^P - a_{jt-1}^P, a_{jt-k}^P - a_{jt-k-1}^P \right) \\
 &= \rho_{aa}^k \text{Var} \left(a^P \right) - \rho_{aa}^{k-1} \text{Var} \left(a^P \right) - \rho_{aa}^{k+1} \text{Var} \left(a^P \right) + \rho_{aa}^k \text{Var} \left(a^P \right) \\
 &= -\rho_{aa}^{k-1} \frac{1 - \rho_{aa}}{1 + \rho_{aa}} \sigma_{aP}^2
 \end{aligned}$$

This gives us the general formula

$$\Sigma_k^a = \begin{cases} \frac{2}{1 + \rho_{aa}} \sigma_{aP}^2 + 2\sigma_{aT}^2 & k = 0 \\ -\frac{1 - \rho_{aa}}{1 + \rho_{aa}} \sigma_{aP}^2 - \sigma_{aT}^2 & k = 1 \\ -\rho_{aa}^{k-1} \frac{1 - \rho_{aa}}{1 + \rho_{aa}} \sigma_{aP}^2 & k > 1 \end{cases} \quad (\text{A21})$$

Demand Similarly, we start by imposing stationarity:

$$(1 - \rho_{bb}^2) \text{Var}(b^P) = \rho_{ba}^2 \text{Var}(a^P) + 2\rho_{ba}\rho_{bb} \text{Cov}(a^P, b^P) + \sigma_{bP}^2$$

$$\text{Cov}(a^P, b^P) = \rho_{aa}\rho_{ba} \text{Var}(a^P) + \rho_{aa}\rho_{bb} \text{Cov}(a^P, b^P)$$

With $\text{Var}(a^P)$ known from earlier steps, these two equations pin down $\text{Var}(b^P)$ and $\text{Cov}(a^P, b^P)$. From here on, we treat the two values as known.

$$k = 0$$

$$\begin{aligned} \Sigma_0^b &= \text{Var}(b_{jt}^P - b_{jt-1}^P + \tau_{jt}^b - \tau_{jt-1}^b) \\ (\text{independence of } \tau^b) &= \text{Var}(b_{jt}^P - b_{jt-1}^P) + \text{Var}(\tau_{jt}^b - \tau_{jt-1}^b) \\ (\text{expand terms}) &= \text{Var}(\rho_{ba}a_{jt-1}^P + (\rho_{bb} - 1)b_{jt-1}^P + \varepsilon_{jt}^b) + 2\sigma_{bT}^2 \\ (\text{expand terms}) &= \rho_{ba}^2 \text{Var}(a^P) + (1 - \rho_{bb})^2 \text{Var}(b^P) + \sigma_{bP}^2 - 2\rho_{ba}(1 - \rho_{bb}) \text{Cov}(a^P, b^P) + 2\sigma_{bT}^2 \end{aligned}$$

For higher-order terms we use the sequential form of b_{jt}^P derived above, whose error terms are independent of the values from $t - k$.

$$k = 1$$

$$\begin{aligned} \Sigma_1^b &= \text{Cov}(b_{jt}^P - b_{jt-1}^P + \tau_{jt}^b - \tau_{jt-1}^b, b_{jt-1}^P - b_{jt-2}^P + \tau_{jt-1}^b - \tau_{jt-2}^b) \\ (\text{independence of } \tau^b) &= \text{Cov}(b_{jt}^P - b_{jt-1}^P, b_{jt-1}^P - b_{jt-2}^P) - \sigma_{bT}^2 \\ (\text{expand terms}) &= \text{Cov}(b_{jt}^P, b_{jt-1}^P) - \text{Var}(b_{jt-1}^P) - \text{Cov}(b_{jt}^P, b_{jt-2}^P) + \text{Cov}(b_{jt-1}^P, b_{jt-2}^P) - \sigma_{bT}^2 \end{aligned}$$

Now we can use the sequential form.

$$\begin{aligned} \text{Cov}(b_{jt}^P, b_{jt-1}^P) &= \text{Cov}(b_{jt-1}^P, b_{jt-2}^P) = \text{Cov}(\rho_{bb}b_{jt-1}^P + Q_1a_{jt-1}^P + \text{error}, b_{jt-1}^P) \\ &= Q_1 \text{Cov}(a^P, b^P) + \rho_{bb} \text{Var}(b^P) \end{aligned}$$

$$\begin{aligned} \text{Cov} \left(b_{jt}^P, b_{jt-2}^P \right) &= \text{Cov} \left(\rho_{bb}^2 b_{jt-2}^P + Q_2 a_{jt-2}^P + \text{error}, b_{jt-2}^P \right) \\ &= Q_2 \text{Cov} \left(a^P, b^P \right) + \rho_{bb}^2 \text{Var} \left(b^P \right) \end{aligned}$$

Plugging in

$$\Sigma_1^b = - (1 - \rho_{bb})^2 \text{Var} \left(b^P \right) + (2Q_1 - Q_2) \text{Cov} \left(a^P, b^P \right) - \sigma_{bT}^2$$

$k > 1$

$$\begin{aligned} \Sigma_k^b &= \text{Cov} \left(b_{jt}^P - b_{jt-1}^P + \tau_{jt}^b - \tau_{jt-1}^b, b_{jt-k}^P - b_{jt-k-1}^P + \tau_{jt-k}^b - \tau_{jt-k-1}^b \right) \\ (\text{independence of } \tau^b) &= \text{Cov} \left(b_{jt}^P - b_{jt-1}^P, b_{jt-k}^P - b_{jt-k-1}^P \right) \\ &= \text{Cov} \left(b_{jt}^P, b_{jt-k}^P \right) - \text{Cov} \left(b_{jt-1}^P, b_{jt-k}^P \right) - \text{Cov} \left(b_{jt}^P, b_{jt-k-1}^P \right) + \text{Cov} \left(b_{jt-1}^P, b_{jt-k-1}^P \right) \\ &= \left(2\rho_{bb}^k - \rho_{bb}^{k-1} - \rho_{bb}^{k+1} \right) \text{Var} \left(b^P \right) + (2Q_k - Q_{k-1} - Q_{k+1}) \text{Cov} \left(a^P, b^P \right) \\ &= -\rho_{bb}^{k-1} (1 - \rho_{bb})^2 \text{Var} \left(b^P \right) + (2Q_k - Q_{k-1} - Q_{k+1}) \text{Cov} \left(a^P, b^P \right) \end{aligned}$$

This gives us the general formula

$$\Sigma_k^b = \begin{cases} \rho_{ba}^2 \text{Var} \left(a^P \right) + (1 - \rho_{bb})^2 \text{Var} \left(b^P \right) + \sigma_{bP}^2 - 2\rho_{ba}(1 - \rho_{bb}) \text{Cov} \left(a^P, b^P \right) + 2\sigma_{bT}^2 & k = 0 \\ - (1 - \rho_{bb})^2 \text{Var} \left(b^P \right) + (2Q_1 - Q_2) \text{Cov} \left(a^P, b^P \right) - \sigma_{bT}^2 & k = 1 \\ -\rho_{bb}^{k-1} (1 - \rho_{bb})^2 \text{Var} \left(b^P \right) + (2Q_k - Q_{k-1} - Q_{k+1}) \text{Cov} \left(a^P, b^P \right) & k > 1 \end{cases} \quad (\text{A22})$$

B.2 Moment Conditions for Elasticity Estimation

Let

$$\begin{aligned} a_{it}^*(\alpha_L) &= \Delta p_{it} - \frac{\alpha_L}{1 + \alpha_L} \Delta s_{it}, \\ b_{it}^*(\sigma) &= \Delta s_{it} + (\sigma - 1) \Delta p_{it}, \end{aligned}$$

and define innovation residuals

$$\begin{aligned} u_{a,it} &= a_{it}^* - \rho_{aa} a_{i,t-1}^*, \\ u_{b,it} &= b_{it}^* - \rho_{ba} a_{i,t-1}^* - \rho_{bb} b_{i,t-1}^*. \end{aligned}$$

Let $Z_{i,t-1}$ denote instruments measurable at $t-1$ (e.g. $L^2 \Delta p_{it}, L^2 \Delta s_{it}, L^3 \Delta p_{it}, L^3 \Delta s_{it}$). The dynamic restrictions require

$$\mathbb{E}[Z_{i,t-1} u_{a,it}] = 0, \quad (\text{A23})$$

$$\mathbb{E}[Z_{i,t-1} u_{b,it}] = 0, \quad (\text{A24})$$

$$\mathbb{E}[u_{a,it} u_{b,it}] = 0. \quad (\text{A25})$$

Equations (A23)–(A24) enforce that the quasi-differenced demand and cost shocks are orthogonal to lagged information, while (A25) imposes no contemporaneous correlation between the structural innovations. Together with the variance restrictions

$$\mathbb{E}[(a_{it}^*)^2] = V_a, \quad (\text{A26})$$

$$\mathbb{E}[(b_{it}^*)^2] = V_b, \quad (\text{A27})$$

(with V_a and V_b as nuisance parameters), this yields five moments to identify $(\sigma, \alpha_L, \rho_{aa}, \rho_{ba}, \rho_{bb})$.

B.3 First Stage: Arellano–Bond Estimation

We first estimate the autoregressive parameters $(\rho_{aa}, \rho_{ba}, \rho_{bb})$ using the Arellano–Bond GMM estimator (Arellano and Bond, 1991). The estimator uses lagged levels as instruments for first differences to address the correlation between the fixed effects and the regressors.

B.3.1 Productivity Persistence

For productivity, we estimate:

$$\Delta a_{jt} = \rho_{aa} \Delta a_{j,t-1} + \Delta u_{jt}^a$$

where $\Delta u_{jt}^a = \Delta \varepsilon_{jt}^a + \Delta \tau_{jt}^a$ combines the persistent and transitory innovations. The moment conditions are:

$$E[a_{j,t-s} \Delta u_{jt}^a] = 0 \quad \text{for } s \geq 2$$

B.3.2 Customer Capital Dynamics

For customer capital, we estimate:

$$\Delta b_{jt} = \rho_{ba}\Delta a_{jt-1} + \rho_{bb}\Delta b_{jt-1} + \Delta u_{jt}^b$$

where $\Delta u_{jt}^b = \Delta \varepsilon_{jt}^b + \Delta \tau_{jt}^b$. The moment conditions are:

$$E[x_{jt-s}\Delta u_{jt}^b] = 0 \quad \text{for } x \in \{a, b\}, s \geq 2$$

B.3.3 Implementation and Results

Because the system is near a unit root in b , no single panel estimator is reliable on its own, and we report the persistence parameters across a bracket of five (Table B1): pooled OLS, the within estimator, the random-effects GLS used in Section 4.2, and the two dynamic-panel GMM estimators, implemented as two-step efficient GMM with lagged levels instrumenting the differenced equations and Windmeijer-corrected standard errors.

TABLE B1: PERSISTENCE PARAMETERS ACROSS PANEL ESTIMATORS

	ρ_{aa}	ρ_{bb}	ρ_{ba}
pooled OLS	0.981 (0.003)	0.981 (0.002)	0.022 (0.002)
firm fixed effects (within)	0.639 (0.015)	0.725 (0.010)	0.019 (0.007)
random-effects GLS (marketing panel)	0.898 (0.007)	0.958 (0.002)	0.032 (0.003)
Arellano–Bond (difference GMM)	0.632 (0.076)	0.963 (0.051)	−0.022 (0.041)
Blundell–Bond (system GMM)	0.930 (0.028)	1.164 (0.029)	−0.021 (0.016)

Notes: Marketing panel (He–Mostrom–Sufi flows, our stock), firms with five consecutive years, firm-clustered standard errors in parentheses.

The ranking $\rho_{bb} > \rho_{aa}$ holds in all five rows. Pooled OLS, which loads the firm effects onto the lag, and the within estimator, which has the short- T downward bias, bracket the random-effects estimate from above and below. Near the unit root the differenced-equation instruments are weak: the Arellano–Bond estimate of ρ_{bb} , 0.963 with a standard error of 0.051, is consistent both with the baseline estimate and with unity, and the Blundell–Bond level moments push it above one. The positive cross-effect is a level-information result: the level estimators put ρ_{ba} between 0.019 and 0.032, while both GMM estimators attenuate it to an insignificant −0.02. Validity of the GMM moment conditions rests on sequential exogeneity of the innovations and no serial correlation in the errors; the Hansen test of overidentifying restrictions ($p = 0.42$) and the test for second-order serial correlation in the differenced residuals ($p = 0.38$) support both. The

random-effects row applies the estimator of Section 4.2 to the marketing panel, where it reads 0.898/0.958/0.032; the imputed sample gives 0.904/0.963/0.033, and the baseline panel of the main text, which observes reported zeros, gives 0.910/0.964/0.033. What matters here is the bracket across estimators rather than any single point. The second stage treats these first-stage estimates as known.

Identification Concerns. We consider four identification concerns.

A. Reverse Causality. The primary identification concern is potential feedback from customer capital to productivity. While our baseline specification assumes productivity evolves independently, firms might adjust their production processes in response to brand-related shocks. An expanded system allows for bidirectional effects:

$$\text{Baseline: } a_{jt} = \rho_{aa}a_{jt-1} + \bar{a}_j + \varepsilon_{jt}^a \quad (\text{A28})$$

$$\text{Alternative: } a_{jt} = \rho_{aa}a_{jt-1} + \rho_{ab}b_{jt-1} + \bar{a}_j + \varepsilon_{jt}^a \quad (\text{A29})$$

The cross-coefficient ρ_{ab} bounds the feedback from customer capital to productivity.

B. Unobserved Heterogeneity. A second concern is that time-varying shocks might simultaneously affect both productivity and customer capital, violating our moment conditions. For example, quality improvements could drive both measures. An instrumental-variables approach leverages industry-level variation:

$$\text{First Stage: } \Delta a_{jt} = \varphi_1 Z_{jt} + e_{jt} \quad (\text{A30})$$

$$\text{Second Stage: } \Delta b_{jt} = \rho_{ba} \Delta \log \tilde{a}_{jt-1} + \rho_{bb} \Delta b_{jt-1} + e'_{jt} \quad (\text{A31})$$

where Z_{jt} includes industry-level shifters of productivity that are plausibly exogenous to firm-specific customer capital.

C. Validity of Lag Instruments. The Arellano–Bond approach relies crucially on the validity of lagged levels as instruments. The Hansen test of overidentifying restrictions and the test for second-order serial correlation, reported above, support the lag instruments.

D. Nonlinear Dynamics. Our linear AR(1) specification may miss important nonlinearities in the evolution of productivity and customer capital. An expanded specification allows for state-dependent parameters:

$$\Delta b_{jt} = \rho_{ba}(s_{jt-1}) \Delta a_{jt-1} + \rho_{bb}(s_{jt-1}) \Delta b_{jt-1} + \Delta \varepsilon_{jt}^b \quad (\text{A32})$$

where the coefficients are flexible functions of the lagged market share.

B.4 Second Stage: GMM Moment Matching

Given these first-stage estimates, we recover the variance parameters $(\sigma_{aT}^2, \sigma_{bT}^2, \sigma_{aP}^2, \sigma_{bP}^2)$ by matching the empirical autocovariances of first differences to the theoretical functions (A21) and (A22) derived in the model-structure section above.

B.4.1 Implementation Details

Key numerical considerations:

- Near unit root in customer capital ($\rho_{bb} \approx 1$) requires careful handling of variance terms
- Regularization in matrix inversions:

$$A = \begin{bmatrix} 1 - \rho_{bb}^2 + 1e - 6 & -\rho_{ba} \\ -\rho_{aa}\rho_{ba} & 1 - \rho_{aa}\rho_{bb} \end{bmatrix}$$

- Treatment of initial conditions

B.4.2 Results

Our second-stage estimates are:

$$\sigma_{aP} = 0.5534 \text{ (Productivity innovation std)}$$

$$\sigma_{aT} = 0.3348 \text{ (Transitory productivity shock std)}$$

$$\sigma_{bP} = 2.1619 \text{ (Customer capital innovation std)}$$

$$\sigma_{bT} = 0.0034 \text{ (Transitory customer-capital shock std)}$$

These components price the permanent layer of the proxies rather than the within-firm innovation, so σ_{bP} is not comparable to the calibrated demand shock $\nu_b = 0.281$, which prices only the within-firm innovation the baseline VAR targets ($\hat{\nu}_b^2 = 0.092$). The estimates reveal three main messages. Customer capital innovations have larger variance than productivity innovations. Productivity has significant transitory components while customer capital is primarily persistent. The model fits productivity autocovariance well but shows some deviation in customer capital autocovariance at longer lags.

B.5 Identification Discussion

The separate identification of variance components comes from a few forces. First, transitory shocks $(\sigma_{aT}^2, \sigma_{bT}^2)$ are primarily identified by $k = 1$ autocovariances. Second, persistent shock variances $(\sigma_{aP}^2, \sigma_{bP}^2)$ are identified by decay rates at higher lags. Third, cross-persistence ρ_{ba} helps identify the relative contribution of productivity versus customer capital shocks.

B.6 Different Selection Criteria

For our main results, we use all firms and do not condition on a balanced panel. Qualitatively, we find very similar results when we condition on a balanced panel.

B.7 Local Projections for Correlations

In this analysis, we examine the relationship between customer capital (b) and cost productivity (a) using local projections. Local projections provide a flexible approach to estimating impulse response functions without imposing the dynamic restrictions inherent in vector autoregressions.

We estimate the following regression for each horizon h :

$$b_{i,t+h} = \beta_0^h + \beta_1^h a_{i,t} + \beta_2^h b_{i,t-1} + \beta_3^h a_{i,t-1} + \beta_4^h b_{i,t-2} + \beta_5^h a_{i,t-2} + \varepsilon_{i,t}^h$$

where $b_{i,t}$ represents customer capital and $a_{i,t}$ represents cost productivity for firm i at time t . The regression is weighted by the firm's share of the product group.

After estimating these local projections, we construct predicted values of customer capital based on the estimated coefficients:

$$\hat{b}_{i,t+h} = \begin{cases} \hat{\beta}_1^h a_{i,t} & \text{if } t = 0 \\ \hat{\beta}_1^h a_{i,t} + \hat{\beta}_2^h \hat{b}_{i,t+h-1} & \text{if } t > 0 \end{cases}$$

The correlation between the predicted customer capital ($\hat{b}_{i,t+1}$) and the realized future cost productivity ($a_{i,t+1}$) is 0.71, indicating a strong positive relationship between the predicted values of customer capital and future cost productivity. This framework allows us to purge the contemporaneous negatively correlated shocks between the two to focus on the dynamics of their relationship consistent with the model.

B.8 Customer-Capital Investment Momentum

The near-unit-root persistence of the customer-capital stock in Section 4.2 is the cumulation of momentum in the underlying investment, which we document with a growth-form distributed lag. We regress the growth of the customer-capital stock on its own lag and on $K+1$ lags of productivity growth,

$$\Delta \log b_{it} = \rho \Delta \log b_{i,t-1} + \sum_{k=0}^K \beta_k \Delta \log \tilde{a}_{i,t-k} + \varepsilon_{it}, \quad (\text{A33})$$

on firms with five consecutive years of data in the merged Compustat/Capital IQ panel, with productivity measured by the revenue-based index $\log \tilde{a}_{it} = \log R_{it} - \bar{\alpha}_B \log b_{it}$.

The own-growth coefficient is positive, $\rho = +0.038$, and stable across the productivity-lag horizon (Table B2). A convex adjustment cost cannot produce this: David and Venkateswaran (2019) show that under quadratic adjustment the conditional autocorrelation of investment growth satisfies $\rho \leq 0$, and physical-capital investment on the same firms mean-reverts. Positive own-autocorrelation in customer-capital investment therefore reflects propagation beyond convex adjustment: a firm that grew its customer base last period keeps growing it, independently of its productivity. Compounded period over period, this momentum sustains the near-unit-root stock persistence ρ_{bb} that drives the dynamic misallocation in Section 4.2. The productivity-growth terms shift from a negative contemporaneous coefficient to a positive one at the three-year lag, consistent with a cross-effect that builds over several years rather than within the year.

TABLE B2: CUSTOMER-CAPITAL GROWTH: DISTRIBUTED-LAG ROBUSTNESS

	$K = 0$	$K = 1$	$K = 2$	$K = 3$
$\Delta \log b_{t-1}$	+0.038 (0.018)	+0.038 (0.018)	+0.037 (0.018)	+0.037 (0.018)
$\Delta \log \tilde{a}_t$	-0.042 (0.027)	-0.042 (0.028)	-0.042 (0.028)	-0.043 (0.028)
$\Delta \log \tilde{a}_{t-1}$		-0.005 (0.017)	-0.004 (0.017)	-0.003 (0.017)
$\Delta \log \tilde{a}_{t-2}$			+0.004 (0.017)	+0.001 (0.017)
$\Delta \log \tilde{a}_{t-3}$				+0.035 (0.014)
R^2	0.004	0.004	0.004	0.006
Observations	5,380			
Firms	1,383			

Notes: Distributed-lag regressions of customer-capital stock-growth on its own lag and $K+1$ lags of productivity-growth: $\Delta \log b_t = \rho \Delta \log b_{t-1} + \sum_{k=0}^K \beta_k \Delta \log \tilde{a}_{t-k} + \varepsilon$. The productivity index $\log \tilde{a}_{it} = \log R_{it} - \alpha_B \log b_{it}$ is the residual from the labor-substituted revenue equation, with α_B estimated by OLS of NAICS3×year-demeaned $\log R$ on demeaned $\log b$ in the broader Capital IQ panel. NAICS3×year fixed effects implemented by demeaning all log levels by industry-year prior to differencing. Sample: firms with five consecutive years of customer capital, physical capital, and $\tilde{a}$ on the merged Compustat/Capital IQ panel; dollar variables deflated to 2017 real dollars. Standard errors clustered at the firm level in parentheses. Under convex adjustment cost alone (David and Venkateswaran 2019), the conditional autocorrelation of investment growth satisfies $\rho \leq 0$; here $\rho > 0$ across all K , robust to productivity-lag horizon.

C Scanner-Data Corroboration

Section 2 measured customer capital from its observable input, marketing expenditure. That measure is directly customer-capital relevant but partial: it captures the demand a firm builds by spending rather than the demand it commands. NielsenIQ Retail Measurement Services (RMS) scanner data report prices and quantities at the barcode level within narrowly defined product groups, so that, under the structural demand system of Hottman et al. (2016), we decompose market shares into cost productivity, markups,

and a residual demand component, and identify that residual with customer capital, the demand state b of the model in Section 3. The two strategies are complementary: Section 2 observes what firms spend to build demand, while here we recover, under a structural assumption, the demand they command from data on prices and quantities. Agreement between the two measures supports the construct: the residual-demand measure shows the same persistence and size-correlation as the marketing-based stock. In product-level data, residual demand explains roughly four times as much as productivity in the cross-sectional variation of firm size, and demand is more persistent than productivity; the gross and net flows of customer capital, through product entry, continuation, and exit, give an independent check on the decay (δ) and marketing-cost (κ) parameters the calibration recovers from the Compustat dynamics (Section 4).

C.1 Data

Residual demand is a useful output-side metric of customer capital: given prices, the demand a firm commands within a product group identifies the stock the model calls b . We measure it with barcode-level data from the Kilts-NielsenIQ Retail Measurement Services, which cover prices and quantities for a broad set of products, brands, and firms across U.S. consumer-product categories, and we merge GS1 company-level information to assign each barcode to a firm (Argente et al., 2024). The product-level detail also lets us decompose the gross and net flows of customer capital.

For each product group k , we observe sales and prices at the firm-year level. Following Hottman et al. (2016), we focus on price and market share variation within product groups to control for differences across categories. Consistent with the model, we focus primarily on the largest firms in each product group; approximately 10 firms per group have a market share above 1% of the national share.

Our final sample includes 423 firms per product group on average, with substantial variation in firm size: the stark difference between mean sales (almost \$6M) and median sales (\$12,000) illustrates the skewness of firm size. The largest firm in a typical product group captures 27% market share, while the second largest firm captures 13%. Market share drops off quickly beyond the top firms; the 10th largest firm captures only a little over 1% share.

C.2 Residual Demand and Firm Size

The structural demand system of [Hottman et al. \(2016\)](#) decomposes the cross-sectional variance of log firm size within a product group into three orthogonal sources: residual demand (the output-side counterpart of customer capital b), cost productivity (a), and markups. On the scanner panel the decomposition puts the bulk of the variation on demand: residual demand accounts for 0.80 of the variance of firm size against 0.20 for productivity (Table C1), roughly four to one. This is the output-side counterpart of Fact 1, where the marketing-based stock tracks size far more closely than productivity does.

TABLE C1: VARIANCE OF FIRM SIZE: RESIDUAL DEMAND VS. PRODUCTIVITY (NIELSEN IQ)

	residual demand b	productivity a
share of the variance of log firm size	0.80	0.20

Notes: Share of the within-product-group cross-sectional variance of log firm size attributed to residual demand versus cost productivity in the [Hottman et al. \(2016\)](#) structural decomposition, NielsenIQ scanner panel, with the markup component netted out before the two shares are formed. Source: RMS NielsenIQ, authors' calculations.

C.3 Relative Persistence of Demand and Productivity

We adapt the static [Hottman et al. \(2016\)](#) demand system to a dynamic setting. In the reduced form, share growth is more persistent than price growth ($\phi_s > \phi_p$), so the demand component of a firm's position decays more slowly than the cost component.

Identification. We combine the static slope conditions of the CES demand system with dynamic orthogonality restrictions in the spirit of [Arellano and Bond \(1991\)](#): observable proxies for productivity and demand changes are built from price and share growth, and their quasi-differenced innovations are required to be orthogonal to lagged information (the full moment conditions are in Appendix B.2). With the substitution elasticity fixed at the value the quantitative analysis uses throughout, $\sigma = 4$

$$\rho_b = \frac{(1 + \xi)\phi_s + \xi(\sigma - 1)\phi_p}{1 + \sigma\xi}, \quad \rho_a = \frac{\xi(\sigma - 1)\phi_s + (1 + \xi)\phi_p}{1 + \sigma\xi}.$$

Demand is more persistent than productivity whenever $\phi_s > \phi_p$ and $\xi(\sigma - 2) < 1$; both hold on the scanner panel. The residual-demand stock is close to a unit root and outlives productivity, $\rho_b = 0.95$ against $\rho_a = 0.90$ (Table C2), the output-side counterpart of Fact 2

and the same ordering the marketing-based stock delivers on the input side.

TABLE C2: STRUCTURAL PERSISTENCE OF DEMAND AND PRODUCTIVITY (NIELSEN IQ)

	demand ρ_b	productivity ρ_a
scanner residual-demand system	0.95	0.90

Notes: Structural persistences from the dynamic [Hottman et al. \(2016\)](#) demand system on the NielsenIQ panel, with $\sigma = 4$ imposed. Source: RMS NielsenIQ, authors' calculations.

C.4 Dynamics of Demand

We now decompose the forces driving demand fluctuations at the firm level. Firm-level demand changes through three channels: product entry, continuation, and exit. We decompose both sales growth and residual demand into these margins.

New product entry explains approximately 23% of the variation in aggregate sales, consistent with [Argente et al. \(2024\)](#). In the model, the two core drivers of demand changes are endogenous investment (new product entry) and natural decay (continuing and exiting products). We group the latter two together and decompose residual demand as

$$\Delta b_{ikt} = \underbrace{\frac{N_{entry}}{N_{ikt}} \cdot b_{entry}}_{\text{entry}} + \underbrace{\frac{N_{cont}}{N_{ikt}} \cdot (b_{cont,t} - b_{cont,t-1}) - \frac{N_{exit}}{N_{ikt,t-1}} \cdot b_{exit}}_{\text{decay}}. \quad (\text{A34})$$

New products require active investment by firms. Given limited consumer attention, entry crowds out existing products. The slopes from projecting the creation and decay flows of customer capital on the lagged stock for the top 10 firms in each product group identify how the gross flows scale with firm customer capital, a check on the decay rate and marketing cost of the calibrated model.

Larger firms create more new product demand in absolute terms but proportionally less: a firm with one log point higher customer capital introduces 0.07 log points more new product demand, so marketing intensity declines with size, as in the Compustat gradient of Section 4. These firms also face higher decay: 0.13 log points per log point of customer capital. The decay slope directly identifies δ ; the creation slope serves as an out-of-sample test for the model's endogenous investment mechanism.

These findings corroborate the marketing-based measure: demand heterogeneity is central to firm size, and it is persistent.